THE SIEVED MIRROR KERNEL $\Phi_X(u)$, THE ZERO-FREE EXPONENTIAL PRIME-GAUGE FACTOR $\mathcal{E}_X(z)$, AND QUADRATIC PRIME-POWER PURITY ON $(X, X^2]$ FOR RIEMANN'S Ξ-FUNCTION

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ABSTRACT. We couple the mother-wavelet decomposition $\Phi(u) = \sum_{m=1}^{\infty} m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$ of Riemann's Jacobi theta kernel with finite-divisor Euler-product sieving for the transverse Wronskian $W(d,t) = \frac{1}{2}\partial_d|\Xi(t - \mathrm{i}d)|^2$. Multiplying $\Xi(z)$ by a self-mirrored Dirichlet polynomial $\mathcal{M}_X(z) = M_X(\frac{1}{2} + \mathrm{i}z)M_X(\frac{1}{2} - \mathrm{i}z)$ corresponds in the u-domain to an even kernel $\Phi_X(u)$ preserving the holomorphic strip $|\mathrm{Im}(u)| < \frac{\pi}{8}$, whereas replacing $M_X(s)$ by the entire, zero-free exponential prime-gauge factor $E_X(s) = \exp(-\sum_{n=2}^{\lfloor X\rfloor} \frac{\Lambda(n)}{\ln n} n^{-s})$ eliminates parasitic mollifier zeros and establishes quadratic prime-power purity $b_N^{\mathcal{E}}(X) \equiv \frac{\Lambda(N)}{\ln N}$ across $\lfloor X\rfloor < N < (\lfloor X\rfloor + 1)^2$. Furthermore, we derive the negative hyperbolic mirror weight $w_n^*(d_0) = -(n^{2d_0} - 1)^{-1}$ that annihilates prime-power oscillations on $d = d_0$ with zero waterbed tail for $n > X$, prove power-law L^2 damping $\frac{X^{-2d}}{2d\ln X}$ and Sub-Selberg closure $\theta_{\mathcal{E}}^{\mathrm{sq}}(d,t) < \frac{1}{2}$, and establish the finite-divisor identity on $[1, N(t)]$ alongside the outer factorization of $\mathcal{E}_{X,\kappa}(z)$.

1. Introduction and Main Results

A foundational technique in the analytic theory of the Riemann zeta function $\zeta(s)$ is the attenuation of Dirichlet-series fluctuations by finite-divisor mollifiers $M_X(s) = \sum_{k\le X} c_k k^{-s}$ approximating $\zeta(s)^{-1} = \sum_{k=1}^{\infty} \mu(k)k^{-s}$, pioneered by Bohr and Landau, developed by Selberg [18], and refined via differential calculus by Levinson [9] and Conrey [1]. When applied to the transverse Wronskian $W(d,t) := \frac{1}{2}\partial_d|\Xi(t - \mathrm{i}d)|^2 = R(d,t)I_t(d,t) - I(d,t)R_t(d,t)$ of Riemann's completed function $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z) = R(d,t) + \mathrm{i}\,I(d,t)$ ($z = t - \mathrm{i}d$, $s = \frac{1}{2} + d + \mathrm{i}t$) [17, 20, 5, 8] or to de Branges Hilbert-space positivity criteria [2, 3], classical polynomial mollifiers face four structural obstacles:

(1) *Parasitic zeros and logarithmic-derivative waterbed tails.* Because a finite Dirichlet polynomial $M_X(s)$ possesses zeros of its own in $\mathbb{C}$, its logarithmic derivative $M_X'(s)/M_X(s)$ places $M_X(s)$ in the denominator, generating an infinite Dirichlet convolution tail for $n > X$ and potential small-denominator spikes.

(2) *Logarithmic versus power-law sieving.* Under smooth polynomial weights $c_k = \mu(k)P(\frac{\ln(X/k)}{\ln X})$ with $P(x) \sim x^r$, low harmonic shells $2 \le N \le X$ are only suppressed logarithmically by $(\ln X)^{-r}$ rather than annihilated identically, while for $N > X$ the sieved coefficients $b_N = \sum_{k|N,\, k\le X} c_k$ can suffer divisor-bound inflation $|b_N| \le d(N)$.

(3) *Conjugate mirror inflation off the critical line.* To preserve real positivity on the critical line $d = 0$, a multiplier must take the self-mirrored form $\mathcal{M}_X(z) = M_X(\frac{1}{2} + \mathrm{i}z)M_X(\frac{1}{2} - \mathrm{i}z)$. At $z = t - \mathrm{i}d$ ($d > 0$), the conjugate factor $M_X(\frac{1}{2} - d - \mathrm{i}t)$ evaluates at $\sigma_{\mathrm{conj}} = \frac{1}{2} - d < \frac{1}{2}$, where individual Dirichlet modes grow as $n^{-(1/2-d)} = n^{-1/2}n^{+d}$.

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(4) *The Selberg barrier* $\theta = \frac{1}{2}$. Covering all active modes $1 \le n \le N(t) := \lfloor\sqrt{t/2\pi}\rfloor$ of the Riemann–Siegel main sum [19, 4] by a linear sieve requires $X \ge N(t)$ ($\theta := \frac{\ln X}{\ln(t/2\pi)} \ge \frac{1}{2}$).

In Paper V of this series [14], following the Wigner phase-space [23, 6], parity-reduced Wronskian, Chladni chamber, and de Branges–Jensen frameworks of Papers I–IV [10, 11, 12, 13], we proved that every mode $\phi_m(u)$ of the Jacobi–Riemann kernel $\Phi(u) = \sum_{m=1}^{\infty} \phi_m(u)$ [7, 16, 22] is an exact logarithmic translation $\phi_m(u) = m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$ of a single universal mother wavelet $\varphi_1(w) = (2\pi^2 e^{9w} - 3\pi e^{5w})\exp(-\pi e^{4w})$. In the present paper (Paper VI), we resolve all four mollifier obstacles by introducing the *Sieved Mirror Kernel* $\Phi_X(u)$ in the u-domain and the entire, zero-free *Exponential Prime-Gauge Factor* $\mathcal{E}_X(z)$. Our main results are consolidated in the following theorem.

Theorem 1.1 (Sieved Mirror Kernel, Zero-Free Prime-Gauge Purity, and Sub-Selberg Closure)**.** *Let* $s = \frac{1}{2} + d + \mathrm{i}t$ *and* $z = t - \mathrm{i}d$ *with* $d \ge 0$ *and* $t > 0$, *let* $N(t) := \lfloor\sqrt{t/2\pi}\rfloor$ *and* $L := \ln(t/2\pi)$, *and let* $\Phi(u) = \sum_{m=1}^{\infty} m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$. *Then the following four structural laws hold:*

(i) ***Sieved Mirror Kernel*** $\Phi_X(u)$***, Strip Preservation*** $|\mathrm{Im}(u)| < \frac{\pi}{8}$***, and Schur-Optimal Filter*** $P_{\mathrm{opt}}(x)$**.** *For any finite real Dirichlet polynomial* $M_X(s) = \sum_{k=1}^{\lfloor X\rfloor} c_k k^{-s}$ *with* $c_1 = 1$, *the self-mirrored product* $\Xi_X(z) := \Xi(z)\mathcal{M}_X(z)$ *with* $\mathcal{M}_X(z) := M_X(\frac{1}{2} + \mathrm{i}z)M_X(\frac{1}{2} - \mathrm{i}z)$ *preserves all zeros of* $\Xi(z)$ *(*$\mathcal{Z}(\Xi) \subseteq \mathcal{Z}(\Xi_X)$*), preserves critical-line mirror symmetry (*$I_X(0,t) \equiv 0$, $\partial_d R_X(0,t) \equiv 0$, $\frac{1}{2}\partial_d|\Xi_X(t - \mathrm{i}d)|^2 = W_X(d,t)$*), and admits the integral representation* $\Xi_X(z) = 8\int_0^{\infty} \Phi_X(u)\cos(2zu)\,\mathrm{d}u$, *where the even* Sieved Mirror Kernel

$$\Phi_X(u) := \sum_{j,k \le X} \frac{c_j c_k}{2\sqrt{jk}} \left[\Phi\Big(u + \frac{1}{2}\ln\frac{j}{k}\Big) + \Phi\Big(u - \frac{1}{2}\ln\frac{j}{k}\Big)\right] \tag{1.1}$$

is holomorphic throughout the full strip $|\mathrm{Im}(u)| < \frac{\pi}{8}$. *Under the forward operator* $\mathcal{T}_X[\Phi](u) = \sum_{N=1}^{\infty} \frac{b_N}{\sqrt{N}}\varphi_1(u + \frac{1}{2}\ln N)$ *with* $b_N = \sum_{k|N,\, k \le X} c_k$, *the mother wavelet* $N = 1$ *remains intact (*$b_1 \equiv 1$*). Within the cubic Conrey–Selberg family* $P(x) = x + a_2 x(1-x) + a_3 x(1-x)^2$, *the unique* 3×3 *Schur-complement minimizer* $(a_2^*, a_3^*) = (+0.81746499, +2.13817017)$ *reduces quadratic cross-term energy* $\mathbf{a}^T\mathbf{Q}\mathbf{a}$ *by a factor of* $31.4\times$ *relative to Conrey's cubic filter* $P_3(x)$.

(ii) ***Zero-Free Exponential Prime-Gauge Factor*** $\mathcal{E}_X(z)$***, Quadratic Purity on*** $(X, X^2]$***, and Negative Hyperbolic Mirror Weight*** $w_n^*(d_0)$**.** *The exponential prime-gauge factor* $E_X(s) := \exp(-\sum_{n=2}^{\lfloor X\rfloor} \frac{\Lambda(n)}{\ln n} n^{-s})$ *and* $\mathcal{E}_X(z) := E_X(\frac{1}{2} + \mathrm{i}z)E_X(\frac{1}{2} - \mathrm{i}z)$ *are entire and zero-free on* $\mathbb{C}$, *so* $\mathcal{Z}(\Xi_{\mathcal{E}_X}) = \mathcal{Z}(\Xi)$ *bijectively. In* $\zeta(s)E_X(s) = \sum_{N=1}^{\infty} b_N^{\mathcal{E}}(X)N^{-s}$, *the coefficients satisfy* $b_1^{\mathcal{E}}(X) = 1$, $b_N^{\mathcal{E}}(X) = 0$ *for all* $2 \le N \le \lfloor X\rfloor$, *strict multiplicativity, universal bounds* $0 \le b_N^{\mathcal{E}}(X_2) \le b_N^{\mathcal{E}}(X_1) \le 1$ *(*$X_2 \ge X_1$*), and* Quadratic Prime-Power Purity*:*

$$b_N^{\mathcal{E}}(X) \equiv \frac{\Lambda(N)}{\ln N} \qquad \text{for all } \lfloor X\rfloor < N < (\lfloor X\rfloor + 1)^2, \tag{1.2}$$

so every composite integer with ≥ 2 *distinct prime factors in* $(X, X^2]$ *vanishes identically. Moreover, in the weighted gauge* $\mathcal{E}_{X,\mathbf{w}}(z)$, *the negative hyperbolic mirror weight* $w_n^*(d_0) := -(n^{2d_0} - 1)^{-1} < 0$ *annihilates the effective prime-power amplitude* $C_n(d_0, w_n^*(d_0)) \equiv 0$ *for all* $2 \le n \le \lfloor X\rfloor$ *on* $d = d_0$ *while introducing zero terms for* $n > X$ *(*0% *waterbed tail).*

(iii) ***Power-Law*** L^2 ***Damping*** $\frac{X^{-2d}}{2d\ln X}$ ***and Universal Sub-Selberg Closure*** $\theta_{\mathcal{E}}^{\mathrm{sq}}(d,t) < \frac{1}{2}$**.** *For any sieve cutoff* $X \in [\lfloor\sqrt{N(t)}\rfloor, N(t)]$ *(*$\theta = \frac{\ln X}{L} \in [\frac{1}{4}, \frac{1}{2}]$*), the entire Riemann–Siegel main sum* $1 \le n \le N(t) < (\lfloor X\rfloor + 1)^2$ *lies inside the quadratic purity window (1.2), replacing logarithmic* $(\ln X)^{-2r}$ *decay by the power-law decay*

$$\mathcal{E}_{\mathrm{diag}}^{\mathcal{E},\mathrm{fwd}}(d,t;X) := \sum_{\lfloor X\rfloor < n \le N(t)} \left(\frac{\Lambda(n)}{\ln n}\right)^2 n^{-(1+2d)} \sim \frac{X^{-2d} - N(t)^{-2d}}{2d\,\ln X} < \frac{X^{-2d}}{2d\,\ln X} \longrightarrow 0, \tag{1.3}$$

and monotonicity together with $Y^{\mathcal{E}}(d,t;N(t))=0$ *implies* $X^{\mathcal{E},\mathrm{sq}}_{\mathrm{krit}}(d,t)<N(t)$ *(*$\theta^{\mathrm{sq}}_{\mathcal{E}}(d,t)<\frac12$*) for all* $d\in(0,\frac12]$ *and all* $t\ge 8\pi$ *(*$N(t)\ge 2$*, with* $\theta^{\mathrm{sq}}_{\mathcal{E}}(d,t)=0<\frac12$ *on* $2\pi<t<8\pi$*).*

(iv) ***Exact Finite-Divisor Identity, Composite-Gap Tail Barrier, and Shifted Outer Factorization.*** *The finite truncation* $M^{\mathcal{E}}_{X,N(t)}(s):=\sum_{k=1}^{N(t)}c^{\mathcal{E}}_k(X)k^{-s}$ *generates the exact same coefficients* $b^{\mathcal{E}}_n(X)$ *on* $1\le n\le N(t)$ *as the infinite series* $E_X(s)$*, while for* $n\ge N(t)+1$*, the bound* $0\le b^{\mathcal{E}}_n(X)\le 1$ *prevents tail inflation and composite gaps* $N(t)+1\le n<q_{\mathrm{next}}(X,t)$ *deepen the Gaussian saddle-point barrier by* $\exp(-\pi(q^2_{\mathrm{next}}-(N(t)+1)^2)\cos 4\eta^*)$*. Finally, for any* $\kappa\ge 0$*, the shifted self-mirrored gauge* $\mathcal{E}_{X,\kappa}(z):=E_X(\frac12+\kappa+\mathrm{i}z)E_X(\frac12+\kappa-\mathrm{i}z)$ *satisfies the exact outer factorization*

$$\left|\Xi_{\mathcal{E}_{X,\kappa}}(t-\mathrm{i}d)\right|^2\equiv\left|E_X(\tfrac12+\kappa-d-\mathrm{i}t)\right|^2\cdot\left|\Xi(t-\mathrm{i}d)\,E_X(\tfrac12+\kappa+d+\mathrm{i}t)\right|^2,\tag{1.4}$$

eliminating conjugate n^{+d} *growth whenever* $\kappa\ge d$*.*

The canonical quantitative constants governing Theorem 1.1 are given explicitly by

$$\begin{aligned} a_2^*&=+0.81746499\ldots, & a_3^*&=+2.13817017\ldots,\\ \mathcal{E}_{\mathrm{quad}}(\mathbf{a}^*)&=2.48425572\times10^{-4}, & \Delta_{\mathrm{gap}}(10^8)&=-64{,}798.48\ \mathrm{dex},\\ N_{\mathrm{comp,min}}(10)&=11\times13=143, & N_{\mathrm{comp,min}}(50)&=53\times59=3127. \end{aligned}\tag{1.5}$$

1.1. **Proof overview.** The proof of Theorem 1.1 is organized around three quantitative and structural ingredients:

(1) *Horizontal Cauchy translation in* $|\mathrm{Im}(u)|<\frac{\pi}{8}$ *and* 3×3 *Schur-complement minimization (Section 2).* Expanding $\mathcal{M}_X(z)$ as a finite cosine series $\sum_{j,k\le X}\frac{c_jc_k}{\sqrt{jk}}\cos(z\ln\frac{j}{k})$, we combine $\cos(2zu)\cos(z\ln\frac{j}{k})$ with the double-exponential decay $|\Phi(u)|=\mathcal{O}(\exp(-\pi\mathrm{e}^{4|\mathrm{Re}(u)|}\cos(4\mathrm{Im}u)))$ to shift the contour horizontally by purely real offsets $\pm\frac12\ln\frac{j}{k}$, preserving the full analyticity strip $|\mathrm{Im}(u)|<\frac{\pi}{8}$. We then project the off-diagonal cross-term gradient onto the three-dimensional Conrey polynomial basis $\{\phi_1,\phi_2,\phi_3\}$ and solve the 2×2 Schur-complement normal equations for (a_2^*,a_3^*).

(2) *Logarithmic-derivative generating functions and negative hyperbolic mirror cancellation (Sections 3 and 4).* Using $\ln\zeta(s)+\ln E_X(s)=\sum_{n>\lfloor X\rfloor}\frac{\Lambda(n)}{\ln n}n^{-s}$ for $\mathrm{Re}(s)>1$, we derive the positive tail recursion $b^{\mathcal{E}}_N(X)\ln N=\sum_{n|N,\,n>X}b^{\mathcal{E}}_{N/n}(X)\Lambda(n)$. Whenever $\lfloor X\rfloor<N<(\lfloor X\rfloor+1)^2$, any divisor $n>X$ of N has quotient $N/n\le\lfloor X\rfloor$, forcing $N/n=1$ and $b^{\mathcal{E}}_N(X)=\frac{\Lambda(N)}{\ln N}$. For $X\ge\lfloor\sqrt{N(t)}\rfloor$, this quadratic purity window covers the entire Riemann–Siegel horizon $[1,N(t)]$, yielding the power-law L^2 bound (1.3) and universal Sub-Selberg closure $\theta^{\mathrm{sq}}_{\mathcal{E}}(d,t)<\frac12$ by partial summation over primes. In the normalized Wronskian $\mathrm{Re}(\xi'_{\mathcal{E}_{X,\mathbf{w}}}/\xi_{\mathcal{E}_{X,\mathbf{w}}})$, equating the forward term $(1-w_n)n^{-(1/2+d_0)}$ against the conjugate mirror term $w_nn^{-(1/2-d_0)}$ uniquely determines $w_n^*(d_0)=-(n^{2d_0}-1)^{-1}<0$.

(3) *Divisor locality on* $[1,N(t)]$ *and shifted outer modulus factorization (Section 5).* Because $k\mid n\implies k\le n$, truncating $E_X(s)$ at $k=N(t)$ leaves $b^{\mathcal{E}}_n(X)$ invariant on $1\le n\le N(t)$, while $0\le b^{\mathcal{E}}_n(X)\le1$ and composite gaps beyond $N(t)$ suppress the evanescent saddle-point tail. Finally, evaluating $\mathcal{E}_{X,\kappa}(z)$ at $z=t-\mathrm{i}d$ separates the conjugate factor $|E_X(\frac12+\kappa-d-\mathrm{i}t)|^2>0$ as a scalar outer multiplier (1.4) with exponent $\sigma_{\mathrm{conj}}=\frac12+\kappa-d\ge\frac12$ whenever $\kappa\ge d$.

1.2. **Notation and conventions.** Consistent with Papers I–V [10, 14], we set $s=\sigma+\mathrm{i}t=\frac12+d+\mathrm{i}t$ and $z=-\mathrm{i}(s-\frac12)=t-\mathrm{i}d$, with completed zeta function $\xi(s)=\frac12s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $\Xi(z):=\xi(\frac12+\mathrm{i}z)$. In the canonical Mellin–Jacobi variable $x=\mathrm{e}^{4u}$, $\Xi(z)$ is given by

$$\Xi(z)=4\int_{-\infty}^{\infty}\Phi(u)\cos(2zu)\,\mathrm{d}u=8\int_0^{\infty}\Phi(u)\cos(2zu)\,\mathrm{d}u,\qquad\Phi(u):=\sum_{m=1}^{\infty}\phi_m(u),\tag{1.6}$$

where $\phi_m(u) = (2\pi^2 m^4 \mathrm{e}^{9u} - 3\pi m^2 \mathrm{e}^{5u}) \exp(-\pi m^2 \mathrm{e}^{4u}) = m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$. If one rescales the integration variable to $\tilde{u} = 2u$ with $\widetilde{\Phi}(\tilde{u}) := 2\Phi(\tilde{u}/2)$, then (1.6) takes the single-frequency form $\Xi(z) = 2\int_0^\infty \widetilde{\Phi}(\tilde{u})\cos(\tilde{u}z)\,\mathrm{d}\tilde{u}$; throughout this paper we retain the canonical coordinate u of (1.6) so that the logarithmic shift in $\phi_m(u)$ and $\Phi_X(u)$ is $\frac{1}{2}\ln m$ and the holomorphic strip is $|\mathrm{Im}(u)| < \frac{\pi}{8}$.

We maintain a strict notational distinction between the four multiplier families studied in this paper:

(a) $M_X(s) := \sum_{k=1}^{\lfloor X\rfloor} c_k k^{-s}$ denotes a finite polynomial Dirichlet mollifier, and $\mathcal{M}_X(z) := M_X(\frac{1}{2} + \mathrm{i}z)M_X(\frac{1}{2} - \mathrm{i}z)$ denotes its self-mirrored counterpart;

(b) $E_X(s) := \exp(-\sum_{n=2}^{\lfloor X\rfloor} \frac{\Lambda(n)}{\ln n} n^{-s}) = \sum_{k=1}^{\infty} c_k^{\mathcal{E}}(X) k^{-s}$ denotes the unweighted exponential prime-gauge factor, and $\mathcal{E}_X(z) := E_X(\frac{1}{2} + \mathrm{i}z)E_X(\frac{1}{2} - \mathrm{i}z)$ denotes its self-mirrored product;

(c) $E_{X,\mathbf{w}}(s) := \exp(-\sum_{n=2}^{\lfloor X\rfloor} w_n \frac{\Lambda(n)}{\ln n} n^{-s})$ and $\mathcal{E}_{X,\mathbf{w}}(z) := E_{X,\mathbf{w}}(\frac{1}{2} + \mathrm{i}z)E_{X,\mathbf{w}}(\frac{1}{2} - \mathrm{i}z)$ denote the weighted exponential prime-gauge factors with real weights $\mathbf{w} = (w_2, \dots, w_{\lfloor X\rfloor})$;

(d) $\mathcal{E}_{X,\kappa}(z) := E_X(\frac{1}{2} + \kappa + \mathrm{i}z)E_X(\frac{1}{2} + \kappa - \mathrm{i}z)$ denotes the shifted self-mirrored prime-gauge factor with real shift parameter $\kappa \geq 0$.

The ordinates of the non-trivial zeros of $\zeta(s)$ on the critical line are denoted by γ_k (with $\gamma_1 = 14.13472514\dots$), whereas Euler's constant is denoted by $\gamma_{\mathrm{E}} = \gamma_0^{\mathrm{St}}$ and the Stieltjes constants [21] by γ_k^{St} ($k \geq 1$). Implied constants in $\mathcal{O}(\cdot)$ and $\ll$ are absolute unless subscripts indicate otherwise.

2. The Sieved Mirror Kernel $\Phi_X(u)$, Holomorphic Strip Preservation $|\mathrm{Im}(u)| < \frac{\pi}{8}$, and 3×3 Schur-Complement Optimization $P_{\mathrm{opt}}(x)$

Let $X > 1$ be a finite sieve cutoff and let $P : [0,1] \to \mathbb{R}$ be a smooth filter function satisfying the boundary conditions $P(0) = 0$ and $P(1) = 1$. Define the finite Dirichlet polynomial

$$M_X(s) := \sum_{k=1}^{\lfloor X\rfloor} \frac{c_k}{k^s}, \qquad c_k := \mu(k)\,P\Big(\frac{\ln(X/k)}{\ln X}\Big) \in \mathbb{R} \qquad (c_1 = \mu(1)P(1) = 1), \tag{2.1}$$

and the self-mirrored multiplier $\mathcal{M}_X(z) := M_X(\frac{1}{2} + \mathrm{i}z)M_X(\frac{1}{2} - \mathrm{i}z)$ with $\Xi_X(z) := \Xi(z)\mathcal{M}_X(z)$.

Lemma 2.1 (Trigonometric Cosine Expansion of $\mathcal{M}_X(z)$)**.** *For all $z = t - \mathrm{i}\delta \in \mathbb{C}$, the self-mirrored multiplier $\mathcal{M}_X(z)$ admits the finite cosine expansion*

$$\mathcal{M}_X(z) = \sum_{j,k\leq X} \frac{c_j c_k}{\sqrt{jk}} \cos\Big(z\ln\frac{j}{k}\Big) = \sum_{k\leq X} \frac{c_k^2}{k} + 2\sum_{1\leq j<k\leq X} \frac{c_j c_k}{\sqrt{jk}} \cos\Big(z\ln\frac{k}{j}\Big). \tag{2.2}$$

Proof. Expanding the finite product gives

$$\mathcal{M}_X(z) = \sum_{j,k\leq X} c_j c_k\, j^{-(1/2+\mathrm{i}z)} k^{-(1/2-\mathrm{i}z)} = \sum_{j,k\leq X} \frac{c_j c_k}{\sqrt{jk}} \exp\Big(-\mathrm{i}z\ln\frac{j}{k}\Big).$$

Since the weight $\frac{c_j c_k}{\sqrt{jk}}$ is invariant under index exchange $j \leftrightarrow k$ whereas $\ln(k/j) = -\ln(j/k)$ reverses sign, symmetrizing each (j,k) and (k,j) pair replaces $\exp(-\mathrm{i}z\ln\frac{j}{k})$ by $\frac{1}{2}\mathrm{e}^{-\mathrm{i}z\ln(j/k)} + \frac{1}{2}\mathrm{e}^{+\mathrm{i}z\ln(j/k)} = \cos(z\ln\frac{j}{k})$. □

Before shifting integration contours in the u-domain, we record the analyticity and double-exponential decay of the Jacobi–Riemann kernel $\Phi(u)$ in horizontal strips. As established in Paper I [10], for $u = \xi + \mathrm{i}\eta$ with $\xi \in \mathbb{R}$ and $|\eta| < \frac{\pi}{8}$, we have $\mathrm{Re}(\pi m^2 \mathrm{e}^{4u}) = \pi m^2 \mathrm{e}^{4\xi}\cos(4\eta) > 0$. By the functional equation of the Jacobi theta function $\theta(x) = \sum_{n\in\mathbb{Z}} \mathrm{e}^{-\pi n^2 x} = x^{-1/2}\theta(x^{-1})$ for $\mathrm{Re}(x) > 0$, $\Phi(-u) = \Phi(u)$ is holomorphic throughout the strip $\mathcal{S}_{\pi/8} := \{u \in \mathbb{C} : |\mathrm{Im}(u)| < \frac{\pi}{8}\}$ and satisfies the uniform double-exponential bound

$$|\Phi(\xi + \mathrm{i}\eta)| \ll \mathrm{e}^{9|\xi|}\exp\Big(-\pi\mathrm{e}^{4|\xi|}\cos(4\eta)\Big) \qquad (|\xi| \to \infty,\ |\eta| \leq \eta_0 < \tfrac{\pi}{8}). \tag{2.3}$$

This rapid decay justifies all contour translations and interchanges of summation and integration in the following theorem.

Theorem 2.2 (Sieved Mirror Kernel and Mother-Wavelet Shell Annihilation)**.** *Let $\mathcal{M}_X(z)$ be defined by (2.2) and $\Xi_X(z) := \Xi(z)\mathcal{M}_X(z) = R_X(d,t) + \mathrm{i}\, I_X(d,t)$ for $z = t - \mathrm{i}d$. Then:*

(a) ***Zero inclusion and critical-line mirror symmetry.*** *$\mathcal{M}_X(z)$ is an entire function on $\mathbb{C}$, so $\mathcal{Z}(\Xi) \subseteq \mathcal{Z}(\Xi_X)$, and at every critical-line zero $\gamma_k \in \mathbb{R}$ of $\Xi(z)$ the parity-reduced Wronskian $\widetilde{W}_X(d,t) := W_X(d,t)/d$ satisfies*

$$\widetilde{W}_X(0,\gamma_k) = \big(\Xi'(\gamma_k)\big)^2 \left|M_X(\tfrac{1}{2} + \mathrm{i}\gamma_k)\right|^4 \geq 0. \tag{2.4}$$

Moreover, $\Xi_X(\bar{z}) = \overline{\Xi_X(z)}$, so $R_X(-d,t) = R_X(d,t)$, $I_X(-d,t) = -I_X(d,t)$, $I_X(0,t) \equiv 0$, $\partial_d R_X(0,t) \equiv 0$, $\Delta R_X = \Delta I_X \equiv 0$, and $\frac{1}{2}\partial_d |\Xi_X(t - \mathrm{i}d)|^2 = W_X(d,t) = R_X \partial_t I_X - I_X \partial_t R_X$.

(b) ***Sieved Mirror Kernel $\Phi_X(u)$ and holomorphic strip preservation.*** *For all $z \in \mathbb{C}$, $\Xi_X(z)$ is the Fourier cosine transform*

$$\Xi_X(z) = 4\int_{-\infty}^{\infty} \Phi_X(u)\cos(2zu)\,\mathrm{d}u = 8\int_0^{\infty} \Phi_X(u)\cos(2zu)\,\mathrm{d}u = 2\int_0^{\infty} \widetilde{\Phi}_X(\tilde{u})\cos(\tilde{u}z)\,\mathrm{d}\tilde{u}, \tag{2.5}$$

where $\widetilde{\Phi}_X(\tilde{u}) := 2\Phi_X(\tilde{u}/2)$ and the Sieved Mirror Kernel *$\Phi_X(u)$ defined by (1.1) is even ($\Phi_X(-u) = \Phi_X(u)$) and preserves the full holomorphic strip $|\mathrm{Im}(u)| < \frac{\pi}{8}$.*

(c) ***Mother-Wavelet shell annihilation.*** *Under the one-sided Dirichlet shift operator $\mathcal{T}_X[\Phi](u)$, the mother-wavelet expansion of $\Phi(u)$ transforms into*

$$\mathcal{T}_X[\Phi](u) := \sum_{k \leq X} \frac{c_k}{\sqrt{k}}\,\Phi\Big(u + \frac{1}{2}\ln k\Big) = \sum_{N=1}^{\infty} \frac{b_N}{\sqrt{N}}\,\varphi_1\Big(u + \frac{1}{2}\ln N\Big), \qquad b_N := \sum_{\substack{k \mid N \\ k \leq X}} c_k, \tag{2.6}$$

where $\zeta(s)M_X(s) = \sum_{N=1}^{\infty} b_N N^{-s}$. In particular, the mother wavelet $N = 1$ remains 100% intact ($b_1 = c_1 \equiv 1$), while for $P_1(x) = x$ all composite integers $2 \leq N \leq X$ with ≥ 2 distinct prime factors satisfy $b_N^{(1)} = \frac{\Lambda(N)}{\ln X} = 0$.

Proof. (a) Each term $k^{-(1/2 \pm \mathrm{i}z)} = \exp(-(1/2 \pm \mathrm{i}z)\ln k)$ is entire on $\mathbb{C}$, so $\mathcal{M}_X(z)$ has no poles in $\mathbb{C}$ and $\Xi(z_0) = 0 \implies \Xi_X(z_0) = 0$. At a critical-line zero $\Xi(\gamma_k) = 0$, the product rule gives $\Xi_X'(\gamma_k) = \Xi'(\gamma_k)\mathcal{M}_X(\gamma_k) = \Xi'(\gamma_k)|M_X(\frac{1}{2} + \mathrm{i}\gamma_k)|^2$, whence $\widetilde{W}_X(0,\gamma_k) = (\Xi_X'(\gamma_k))^2$ yields (2.4). Because $c_k \in \mathbb{R}$, Lemma 2.1 and $\cos(\bar{w}) = \overline{\cos w}$ imply $\mathcal{M}_X(\bar{z}) = \overline{\mathcal{M}_X(z)}$. Combining this with $\Xi(\bar{z}) = \overline{\Xi(z)}$ gives $\Xi_X(t + \mathrm{i}d) = \overline{\Xi_X(t - \mathrm{i}d)}$, which forces $R_X(-d,t) = R_X(d,t)$ and $I_X(-d,t) = -I_X(d,t)$. Holomorphicity in $z = t - \mathrm{i}d$ implies $\partial_d R_X = \partial_t I_X$ and $\partial_d I_X = -\partial_t R_X$, yielding harmonicity and $\frac{1}{2}\partial_d(R_X^2 + I_X^2) = R_X\partial_t I_X - I_X \partial_t R_X = W_X(d,t)$.

(b) Multiplying $\Xi(z) = 4\int_{-\infty}^{\infty} \Phi(u)\cos(2zu)\,\mathrm{d}u$ by (2.2) and applying the trigonometric identity

$$\cos(2zu)\cos\Big(z\ln\frac{j}{k}\Big) = \frac{1}{2}\left[\cos\Big(2z\Big(u - \frac{1}{2}\ln\frac{j}{k}\Big)\Big) + \cos\Big(2z\Big(u + \frac{1}{2}\ln\frac{j}{k}\Big)\Big)\right],$$

the translation invariance of Lebesgue integration on $\mathbb{R}$ ($u \mp \frac{1}{2}\ln(j/k) \mapsto u$, justified for all $z \in \mathbb{C}$ by the double-exponential bound (2.3)) yields (2.5) with $\Phi_X(u)$ given by (1.1). Because every shift $\pm\frac{1}{2}\ln(j/k) \in \mathbb{R}$ is purely real, $\mathrm{Im}(u \pm \frac{1}{2}\ln\frac{j}{k}) = \mathrm{Im}(u)$, so the singularities of $\Phi(u \pm \frac{1}{2}\ln\frac{j}{k})$ remain on the horizontal lines $\mathrm{Im}(u) = \pm\frac{\pi}{8}$. Hence $\Phi_X(u)$ is even and holomorphic throughout $|\mathrm{Im}(u)| < \frac{\pi}{8}$.

(c) Substituting $\Phi(u + \frac{1}{2}\ln k) = \sum_{m=1}^{\infty} m^{-1/2}\varphi_1(u + \frac{1}{2}\ln k + \frac{1}{2}\ln m)$ into $\mathcal{T}_X[\Phi](u)$ and grouping the absolutely convergent double series (justified by (2.3)) by the product index $N = km$ gives $\frac{1}{\sqrt{k}}\frac{1}{\sqrt{m}} = \frac{1}{\sqrt{N}}$ and $\frac{1}{2}\ln k + \frac{1}{2}\ln m = \frac{1}{2}\ln N$, proving (2.6). For $N = 1$, only $k = 1$ divides 1, so $b_1 = c_1 = 1$. For Selberg's linear filter $P_1(x) = x = 1 - \frac{\ln k}{\ln X}$ and $2 \leq N \leq X$, the classical identities $\sum_{k \mid N} \mu(k) = 0$ and $-\sum_{k \mid N} \mu(k)\ln k = \Lambda(N)$ give $b_N^{(1)} = \frac{\Lambda(N)}{\ln X}$. □

We now show how the higher-degree polynomial coefficients of $P(x)$ can be chosen via a positive semi-definite quadratic form to minimize off-diagonal cross-term interference between the mother wavelet ($m = 1$) and the satellite shells $2 \le n \le N_{\max}$.

Proposition 2.3 (Quadratic Cross-Term Form and 3×3 Schur-Complement Filter $P_{\mathrm{opt}}(x)$)**.** *Consider the cubic Conrey–Selberg filter family satisfying $P(0) = 0$ and $P(1) = 1$:*

$$P(x) = \phi_1(x) + a_2\,\phi_2(x) + a_3\,\phi_3(x), \qquad \phi_1(x) := x, \quad \phi_2(x) := x(1-x), \quad \phi_3(x) := x(1-x)^2, \tag{2.7}$$

parameterized by $\mathbf{a} := (1, a_2, a_3)^T \in \{1\}\times\mathbb{R}^2$. Then the sieved coefficients $b_n(\mathbf{a}) = \sum_{\alpha=1}^{3} a_\alpha b_n^{(\alpha)}$ are affine-linear in (a_2, a_3) with $b_1(\mathbf{a}) \equiv 1$. For any compact spectral interval $[T_1, T_2]$ with active horizon $N_{\max} = N(T_2)$ and mother-wavelet cross-waveforms $\Psi_\alpha(d,t) := \sum_{n=2}^{N_{\max}} b_n^{(\alpha)} K_{1,n}(d,t)$ (where $K_{1,n}(d,t) := \frac{4}{\sqrt{n}}[L_{1,n}^{+}\sinh(dL_{1,n}^{+})\cos(t\ln n) + (\ln n)\sinh(d\ln n)\cos(2\vartheta(t) - t\ln n)]$ with $L_{1,n}^{+} := \ln(\frac{t}{2\pi n})$ is the mother-wavelet cross-kernel of Paper V [14]), the mean-square cross-term gradient energy is the 3×3 quadratic form

$$\mathcal{E}_{\mathrm{quad}}(\mathbf{a}) := \frac{1}{T_2 - T_1}\int_{T_1}^{T_2}\left(\sum_{\alpha=1}^{3} a_\alpha\,\Psi_\alpha(d,t)\right)^2 \mathrm{d}t = \mathbf{a}^T\mathbf{Q}\,\mathbf{a}, \quad Q_{\alpha\beta} := \frac{1}{T_2 - T_1}\int_{T_1}^{T_2}\Psi_\alpha(d,t)\,\Psi_\beta(d,t)\,\mathrm{d}t. \tag{2.8}$$

Whenever the 2×2 principal block $\mathbf{Q}_{22} := (Q_{\alpha\beta})_{2\le\alpha,\beta\le 3}$ is positive definite, $\mathcal{E}_{\mathrm{quad}}(\mathbf{a})$ possesses the unique global minimizer and Schur-complement minimum value

$$\begin{pmatrix} a_2^* \\ a_3^* \end{pmatrix} = -\mathbf{Q}_{22}^{-1}\begin{pmatrix} Q_{21} \\ Q_{31} \end{pmatrix}, \qquad \mathcal{E}_{\mathrm{quad}}(\mathbf{a}^*) = Q_{11} - \begin{pmatrix} Q_{12} & Q_{13} \end{pmatrix}\mathbf{Q}_{22}^{-1}\begin{pmatrix} Q_{21} \\ Q_{31} \end{pmatrix}. \tag{2.9}$$

On $[T_1, T_2] = [14, 300]$ ($N_{\max} = 6$, $d = 0.20$, $X = 50$), equation (2.9) gives $(a_2^*, a_3^*) = (+0.81746499, +2.13817017)$ and $\mathcal{E}_{\mathrm{quad}}(\mathbf{a}^*) = 2.48425572\times 10^{-4}$, which is $31.41\times$ smaller than Conrey's cubic filter $P_3(x)$ ($\mathbf{a} = (1,1,1)^T$, $\mathcal{E}_{\mathrm{quad}} = 7.8021\times 10^{-3}$) and $2605.7\times$ smaller (99.9616% reduction) than Selberg's linear filter $P_1(x)$ ($\mathbf{a} = (1,0,0)^T$, $\mathcal{E}_{\mathrm{quad}} = 0.647448$).

Proof. Because $\phi_1(1) = 1$ and $\phi_2(1) = \phi_3(1) = 0$, we have $c_1^{(1)} = 1$ and $c_1^{(2)} = c_1^{(3)} = 0$, so $b_1(\mathbf{a}) = 1 + 0\cdot a_2 + 0\cdot a_3 \equiv 1$. By linearity of the finite divisor sum $b_n(\mathbf{a}) = \sum_{k|n,\,k\le X}\sum_{\alpha=1}^{3} a_\alpha c_k^{(\alpha)}$, the cross-term gradient $\sum_{n=2}^{N_{\max}} b_n(\mathbf{a})K_{1,n}(d,t)$ equals $\sum_{\alpha=1}^{3} a_\alpha\Psi_\alpha(d,t)$, whose $L^2([T_1,T_2])$ inner product is (2.8). Writing $\mathbf{a} = (1,\tilde{\mathbf{a}})^T$ with $\tilde{\mathbf{a}} = (a_2,a_3)^T$ and $\mathbf{q}_1 = (Q_{21},Q_{31})^T$, completion of the square gives

$$\mathbf{a}^T\mathbf{Q}\,\mathbf{a} = Q_{11} + 2\mathbf{q}_1^T\tilde{\mathbf{a}} + \tilde{\mathbf{a}}^T\mathbf{Q}_{22}\tilde{\mathbf{a}} = \left(\tilde{\mathbf{a}} + \mathbf{Q}_{22}^{-1}\mathbf{q}_1\right)^T\mathbf{Q}_{22}\left(\tilde{\mathbf{a}} + \mathbf{Q}_{22}^{-1}\mathbf{q}_1\right) + \left(Q_{11} - \mathbf{q}_1^T\mathbf{Q}_{22}^{-1}\mathbf{q}_1\right),$$

establishing (2.9). The explicit Gram matrix entries and interval-arithmetic certification are detailed in Appendix A.1. □

3. The Zero-Free Exponential Prime-Gauge Factor $\mathcal{E}_X(z)$, Quadratic Prime-Power Purity on $(X, X^2]$, and Negative Hyperbolic Mirror Annihilation $w_n^*(d_0)$

While the polynomial mollifier $\mathcal{M}_X(z)$ is pole-free, $M_X(s)$ can vanish in $\mathbb{C}$ and leaves non-zero residual prime shells $b_N \ne 0$ for $2 \le N \le X$. To eliminate both parasitic mollifier zeros and all residual shells up to X, we exponentiate the truncated logarithmic Euler product:

$$E_X(s) := \exp\left(-\sum_{n=2}^{\lfloor X\rfloor}\frac{\Lambda(n)}{\ln n}\,n^{-s}\right) = \sum_{k=1}^{\infty} c_k^{\mathcal{E}}(X)\,k^{-s}, \qquad \mathcal{E}_X(z) := E_X\left(\frac{1}{2} + \mathrm{i}z\right)E_X\left(\frac{1}{2} - \mathrm{i}z\right). \tag{3.1}$$

Because the exponent $\mathcal{P}_X(z) := -2\sum_{n=2}^{\lfloor X\rfloor}\frac{\Lambda(n)}{(\ln n)\sqrt{n}}\cos(z\ln n)$ is a finite sum of entire cosine functions, $\mathcal{E}_X(z) = \exp(\mathcal{P}_X(z))$ has *zero poles and zero parasitic zeros anywhere in* $\mathbb{C}$, so $\Xi_{\mathcal{E}_X}(z) := \Xi(z)\mathcal{E}_X(z)$ satisfies $\mathcal{Z}(\Xi_{\mathcal{E}_X}) = \mathcal{Z}(\Xi)$ bijectively with identical multiplicities.

TABLE 1. Comparison of sieved shell coefficients b_N ($1 \le N \le 9$) and cross-term suppression metrics on $t \in [14, 300]$ ($d = 0.20, X = 50$), certified in `mpmath` (`mp.dps = 50`) and `mpmath.iv` (`iv.dps = 35`).

Quantity / Metric	Unsieved	Selberg $P_1(x)$	Conrey $P_2(x)$	Conrey $P_3(x)$	Schur $P_{\mathrm{opt}}(x)$	Exp-Gauge E_X
Parameter pair (a_2, a_3)	—	$(0, 0)$	$(1, 0)$	$(1, 1)$	$(0.8175, 2.1382)$	Log-Euler
Mother Wavelet b_1 ($N = 1$)	$+1.000000$	$+1.000000$	$+1.000000$	$+1.000000$	$+1.000000$	$+1.000000$
Prime shell b_2 ($N = 2$)	$+1.000000$	$+0.177184$	$+0.031394$	$+0.005563$	$+0.002774$	0.000000
Prime shell b_3 ($N = 3$)	$+1.000000$	$+0.280830$	$+0.078865$	$+0.022148$	-0.005541	0.000000
Prime-power shell b_4 (2^2)	$+1.000000$	$+0.177184$	$+0.031394$	$+0.005563$	$+0.002774$	0.000000
Prime shell b_5 ($N = 5$)	$+1.000000$	$+0.411408$	$+0.169257$	$+0.069634$	$+0.000447$	0.000000
Composite shell b_6 ($2 \cdot 3$)	$+1.000000$	0.000000	-0.099517	-0.068370	-0.014754	0.000000
Prime shell b_7 ($N = 7$)	$+1.000000$	$+0.497418$	$+0.247425$	$+0.123073$	$+0.027173$	0.000000
Quadratic energy $\mathbf{a}^T\mathbf{Q}\mathbf{a}$	—	0.647448	0.054728	0.007802	0.000248	0.000000
Full RS cross-term RMS$\lvert\partial_d \mathcal{C}_N\rvert$	4.150934	0.830237	0.238616	0.090088	0.016084	0.000000
Stability $\min_t(W_X/D_{\mathrm{diag}})$	$+0.019086$	$+0.655414$	$+0.862846$	$+0.943060$	$+0.992231$	$+1.000000$

Theorem 3.1 (Number-Theoretic Characterization of $b_N^{\mathcal{E}}(X)$ and Quadratic Prime-Power Purity Window)**.** *Let $X \ge 1$ and define the sieved Dirichlet coefficients $b_N^{\mathcal{E}}(X)$ for $\mathrm{Re}(s) > 1$ by*

$$\zeta(s)\, E_X(s) = \exp\left(\sum_{n > \lfloor X \rfloor} \frac{\Lambda(n)}{\ln n}\, n^{-s} \right) = \sum_{N=1}^{\infty} b_N^{\mathcal{E}}(X)\, N^{-s}. \tag{3.2}$$

Then the following four arithmetic laws hold:

(i) ***Exact Möbius recursion and shell annihilation up to X.*** *The coefficients $c_k^{\mathcal{E}}(X)$ of $E_X(s)$ satisfy $c_1^{\mathcal{E}}(X) = 1$, $c_k^{\mathcal{E}}(X) \ln k = -\sum_{n \mid k,\, 2 \le n \le \lfloor X \rfloor} c_{k/n}^{\mathcal{E}}(X) \Lambda(n)$ for $k \ge 2$, and $c_k^{\mathcal{E}}(X) \equiv \mu(k)$ for all $1 \le k \le \lfloor X \rfloor$. Simultaneously, $b_N^{\mathcal{E}}(X)$ obeys the positive tail recursion*

$$b_1^{\mathcal{E}}(X) = 1, \qquad b_N^{\mathcal{E}}(X)\, \ln N = \sum_{\substack{n \mid N \\ n > \lfloor X \rfloor}} b_{N/n}^{\mathcal{E}}(X)\, \Lambda(n) \qquad (N \ge 2), \tag{3.3}$$

which implies $b_N^{\mathcal{E}}(X) = 0$ identically for all $2 \le N \le \lfloor X \rfloor$.

(ii) ***Strict multiplicativity and unitary prime-power support.*** *$b_N^{\mathcal{E}}(X)$ is multiplicative: $b_{MN}^{\mathcal{E}}(X) = b_M^{\mathcal{E}}(X) b_N^{\mathcal{E}}(X)$ whenever $\gcd(M, N) = 1$. For an arbitrary integer with canonical prime factorization $N = \prod_{i=1}^{r} p_i^{\alpha_i}$,*

$$b_N^{\mathcal{E}}(X) = \prod_{i=1}^{r} b_{p_i^{\alpha_i}}^{\mathcal{E}}(X) > 0 \iff \text{every unitary prime-power divisor } p_i^{\alpha_i} \parallel N \text{ satisfies } p_i^{\alpha_i} > X. \tag{3.4}$$

(iii) ***Universal 0% overshoot and monotonicity in X.*** *For all $N \ge 1$ and all $X_2 \ge X_1 \ge 1$,*

$$0 \le b_N^{\mathcal{E}}(X_2) \le b_N^{\mathcal{E}}(X_1) \le b_N^{\mathcal{E}}(1) = 1. \tag{3.5}$$

(iv) ***Quadratic Prime-Power Purity on $\lfloor X \rfloor < N < (\lfloor X \rfloor + 1)^2$.*** *Throughout the entire quadratic window $\lfloor X \rfloor < N < (\lfloor X \rfloor + 1)^2$ (and in particular on $(X, X^2]$),*

$$b_N^{\mathcal{E}}(X) \equiv \frac{\Lambda(N)}{\ln N} = \begin{cases} 1 & \text{if } N = p \text{ is prime,} \\ 1/k & \text{if } N = p^k \text{ is a higher prime power } (k \ge 2), \\ 0 & \text{if } N \text{ is composite with } \ge 2 \text{ distinct prime factors.} \end{cases} \tag{3.6}$$

Consequently, the smallest composite integer with ≥ 2 distinct prime factors having $b_N^{\mathcal{E}}(X) > 0$ is $N_{\mathrm{comp,min}}(X) = q_1(X) q_2(X) \ge (\lfloor X \rfloor + 1)(\lfloor X \rfloor + 2) > X^2$, where $q_1(X) < q_2(X)$ are the two smallest coprime prime powers exceeding X.

Proof. (i) For $\mathrm{Re}(s) > 1$, the Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ has logarithmic series $\ln \zeta(s) = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{\ln n} n^{-s}$. Adding $\ln E_X(s) = -\sum_{n=2}^{\lfloor X \rfloor} \frac{\Lambda(n)}{\ln n} n^{-s}$ cancels all terms $2 \le n \le \lfloor X \rfloor$ and yields (3.2). Differentiating $E_X(s)$ and $\zeta(s)E_X(s)$ gives $-E_X'(s) = E_X(s) \sum_{n=2}^{\lfloor X \rfloor} \Lambda(n) n^{-s}$ and $-(\zeta E_X)'(s) = (\zeta E_X)(s) \sum_{n > \lfloor X \rfloor} \Lambda(n) n^{-s}$; equating coefficients of k^{-s} and N^{-s} yields the recursion for $c_k^{\mathcal{E}}(X)$ and (3.3). For $1 \le k \le \lfloor X \rfloor$, every divisor $n \mid k$ satisfies $n \le k \le \lfloor X \rfloor$, so $c_k^{\mathcal{E}}(X)$ obeys the exact same recursion $\mu(k) \ln k = -\sum_{n \mid k} \mu(k/n) \Lambda(n)$ as the Möbius function, proving $c_k^{\mathcal{E}}(X) = \mu(k)$ by induction. For $2 \le N \le \lfloor X \rfloor$, the sum in (3.3) is empty, so $b_N^{\mathcal{E}}(X) = 0$.

(ii) Grouping the exponent $R_{>X}(s) := \sum_{n > \lfloor X \rfloor} \frac{\Lambda(n)}{\ln n} n^{-s}$ by prime base p with threshold exponent $k_p(X) := \lfloor \log_p X \rfloor + 1 = \min\{k \ge 1 : p^k > X\}$ gives the Euler-type factorization

$$\zeta(s)\, E_X(s) = \prod_p \exp\left(\sum_{k=k_p(X)}^{\infty} \frac{p^{-ks}}{k} \right) = \prod_p \left(1 + \sum_{\alpha=1}^{\infty} b_{p^\alpha}^{\mathcal{E}}(X)\, p^{-\alpha s} \right).$$

Because factors indexed by distinct primes p involve coprime Dirichlet frequencies $p^{-\alpha s}$, $b_N^{\mathcal{E}}(X)$ is multiplicative. For a single prime power p^α, the series $\sum_{k \ge k_p(X)} \frac{p^{-ks}}{k}$ starts at degree $k_p(X)$, so $b_{p^\alpha}^{\mathcal{E}}(X) = 0$ if $\alpha < k_p(X)$ (i.e., $p^\alpha \le X$) and $b_{p^\alpha}^{\mathcal{E}}(X) \ge \frac{1}{\alpha} > 0$ if $\alpha \ge k_p(X)$ (i.e., $p^\alpha > X$). Multiplicativity over $p_i^{\alpha_i} \parallel N$ immediately yields (3.4).

(iii) For $X_2 \ge X_1 \ge 1$, write $R_{>X_1}(s) = R_{>X_2}(s) + \Delta(s)$ with $\Delta(s) := \sum_{\lfloor X_1 \rfloor < n \le \lfloor X_2 \rfloor} \frac{\Lambda(n)}{\ln n} n^{-s}$. Since both $R_{>X_2}(s)$ and $\Delta(s)$ have non-negative Dirichlet coefficients, expanding $\exp(R_{>X_1}(s)) = \exp(R_{>X_2}(s)) \sum_{m=0}^{\infty} \frac{1}{m!} \Delta(s)^m$ shows coefficientwise that $0 \le b_N^{\mathcal{E}}(X_2) \le b_N^{\mathcal{E}}(X_1)$. Taking $X_1 = 1$ gives $\exp(R_{>1}(s)) = \zeta(s) = \sum_{N=1}^{\infty} 1 \cdot N^{-s}$, so $b_N^{\mathcal{E}}(1) = 1$, establishing (3.5).

(iv) Let $\lfloor X \rfloor < N < (\lfloor X \rfloor + 1)^2$. In the recursion (3.3), every divisor $n \mid N$ satisfying $n \ge \lfloor X \rfloor + 1$ has complementary quotient $m := N/n < \frac{(\lfloor X \rfloor + 1)^2}{\lfloor X \rfloor + 1} = \lfloor X \rfloor + 1$, so $1 \le m \le \lfloor X \rfloor$. By part (i), $b_m^{\mathcal{E}}(X) = 0$ for all $2 \le m \le \lfloor X \rfloor$, so only $m = 1$ ($n = N$) contributes to (3.3), yielding $b_N^{\mathcal{E}}(X) \ln N = b_1^{\mathcal{E}}(X) \Lambda(N) = \Lambda(N)$ and proving (3.6). □

We next resolve Open Problem 7 (the pointwise waterbed and mirror-inflation effect in the normalized Wronskian). Consider the weighted exponential prime-gauge factor with real weights $\mathbf{w} = (w_2, \dots, w_{\lfloor X \rfloor}) \in \mathbb{R}^{\lfloor X \rfloor - 1}$:

$$E_{X,\mathbf{w}}(s) := \exp\left(-\sum_{n=2}^{\lfloor X \rfloor} w_n \frac{\Lambda(n)}{\ln n} n^{-s} \right), \quad \mathcal{E}_{X,\mathbf{w}}(z) := E_{X,\mathbf{w}}\left(\tfrac{1}{2} + \mathrm{i}z\right) E_{X,\mathbf{w}}\left(\tfrac{1}{2} - \mathrm{i}z\right) = \exp\big(\mathcal{P}_{X,\mathbf{w}}(z)\big), \tag{3.7}$$

where $\mathcal{P}_{X,\mathbf{w}}(z) = -2 \sum_{n=2}^{\lfloor X \rfloor} w_n \frac{\Lambda(n)}{(\ln n)\sqrt{n}} \cos(z \ln n)$, and set $\Xi_{\mathcal{E}_{X,\mathbf{w}}}(z) := \Xi(z) \mathcal{E}_{X,\mathbf{w}}(z)$.

Theorem 3.2 (Exact Negative Hyperbolic Mirror Weight $w_n^*(d_0) = -(n^{2d_0} - 1)^{-1}$ Without Waterbed Tail)**.** *For all $(d,t) \in \mathbb{R}^2$ outside the zeros of Ξ, the normalized transverse Wronskian of $\Xi_{\mathcal{E}_{X,\mathbf{w}}}(z)$ equals $\mathrm{Re}(\xi'(s)/\xi(s)) + \partial_d \mathrm{Re}\, \mathcal{P}_{X,\mathbf{w}}(t - \mathrm{i}d)$ and satisfies the exact additive identity*

$$\frac{W_{\mathcal{E}_{X,\mathbf{w}}}(d,t)}{\left|\Xi_{\mathcal{E}_{X,\mathbf{w}}}(t - \mathrm{i}d)\right|^2} = \mathrm{Re}\left(\frac{\xi'(s)}{\xi(s)}\right) - 2 \sum_{n=2}^{\lfloor X \rfloor} w_n \frac{\Lambda(n)}{\sqrt{n}} \sinh(d \ln n) \cos(t \ln n). \tag{3.8}$$

Combined with the Dirichlet series $-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=2}^{\infty} \Lambda(n) n^{-s}$ and the Archimedean digamma drift $\mathcal{A}(d,t) := \mathrm{Re}(\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \ln \pi + \frac{1}{2} \psi(\frac{s}{2}))$, the effective amplitude of the n-th prime power ($2 \le n \le \lfloor X \rfloor$) is

$$C_n(d, w_n) := (1 - w_n)\, n^{-(1/2+d)} + w_n\, n^{-(1/2-d)} = n^{-(1/2+d)} + 2 w_n\, n^{-1/2} \sinh(d \ln n). \tag{3.9}$$

TABLE 2. Arithmetic structure of $b_N^{\mathcal{E}}(10)$ across $1 \le N \le 176$ (Theorem 3.1) and interval-certified annihilation $C_n(d_0, w_n^*(d_0)) = 0$ at $d_0 = 0.20$ (Theorem 3.2), computed at `mpmath.mp.dps = 50` and `mpmath.iv.dps = 35`.

Theorem 3.1: $b_N^{\mathcal{E}}(10)$ in $\zeta(s)E_{10}(s)$				Theorem 3.2: $w_n^*(d_0)$ at $d_0 = 0.20$				
Shell N	Factorization	$b_N^{\mathcal{E}}(10)$	Purity Regime	n	$n^{-(1/2+d_0)}$	$n^{-(1/2-d_0)}$	$w_n^*(0.20)$	iv Width of C_n
1	1	1.000000	Mother Wavelet	2	0.615572	0.812252	−3.129813	6.92×10^{-35}
$2 \dots 10$	$p \le 10$ / comp.	0.000000	Sieved $N \le X$	3	0.463461	0.719223	−1.812101	3.46×10^{-35}
11, 13	Prime > 10	1.000000	$\Lambda(p)/\ln p = 1$	4	0.378929	0.659754	−1.349344	1.73×10^{-35}
12, 14, 15	$2^2{\cdot}3, 2{\cdot}7, 3{\cdot}5$	0.000000	Unitary $p^\alpha \le 10$	5	0.324136	0.617034	−1.106617	1.73×10^{-35}
16	$2^4 > 10$	0.250000	$\Lambda(2^4)/\ln 16 = 1/4$	7	0.256065	0.556793	−0.848964	8.65×10^{-36}
22, 100	$2{\cdot}11, 2^2{\cdot}5^2$	0.000000	$2^1, 2^2 \parallel N \le 10$	11	0.186607	0.486968	−0.621057	8.65×10^{-36}
25, 27	$5^2, 3^3 > 10$	0.500, 0.333	$\Lambda(p^k)/\ln p^k = 1/k$	16	0.143587	0.435275	−0.492263	4.33×10^{-36}
121	$11^2 > (10+1)^2 - 1$	1.000000	$\frac{1}{2} + \frac{1}{2!}1^2 = 1$	25	0.105061	0.380731	−0.381112	4.33×10^{-36}
143	11×13	1.000000	First comp. $> X^2$	29	0.094694	0.364150	−0.351541	4.33×10^{-36}

For any target transverse line $d_0 > 0$, choosing the negative hyperbolic mirror weight

$$\boxed{w_n^*(d_0) := -\frac{n^{-d_0}}{n^{+d_0} - n^{-d_0}} = -\frac{1}{n^{2d_0} - 1} < 0 \qquad (\forall\, d_0 > 0,\ 2 \le n \le \lfloor X \rfloor)} \tag{3.10}$$

annihilates $C_n(d_0, w_n^*(d_0)) \equiv 0$ identically for all $2 \le n \le \lfloor X \rfloor$ on the line $d = d_0$, while introducing zero terms for $n > X$ (0% convolution/waterbed tail) *in the logarithmic derivative:*

$$\frac{W_{\mathcal{E}_{X,\mathbf{w}^*(d_0)}}(d_0, t)}{\left|\Xi_{\mathcal{E}_{X,\mathbf{w}^*(d_0)}}(t - \mathrm{i}d_0)\right|^2} = \mathrm{Re}\left(\frac{\xi'(\frac{1}{2} + d_0 + \mathrm{i}t)}{\xi(\frac{1}{2} + d_0 + \mathrm{i}t)}\right) + \sum_{n=2}^{\lfloor X \rfloor} \frac{\Lambda(n)}{n^{1/2+d_0}} \cos(t \ln n) \qquad (\forall\, t \in \mathbb{R}). \tag{3.11}$$

Proof. Since $\frac{1}{2}\partial_d \ln |\Xi_{\mathcal{E}_{X,\mathbf{w}}}(t - \mathrm{i}d)|^2 = W_{\mathcal{E}_{X,\mathbf{w}}}(d, t)/|\Xi_{\mathcal{E}_{X,\mathbf{w}}}(t - \mathrm{i}d)|^2$ and $\ln |\Xi_{\mathcal{E}_{X,\mathbf{w}}}| = \ln |\xi(s)| + \mathrm{Re}\, \mathcal{P}_{X,\mathbf{w}}(t - \mathrm{i}d)$, differentiating $\mathrm{Re}\cos((t - \mathrm{i}d)\ln n) = \cosh(d \ln n)\cos(t \ln n)$ with respect to d multiplies by $\ln n \sinh(d \ln n)$, canceling $(\ln n)^{-1}$ and yielding (3.8). Writing $2n^{-1/2}\sinh(d \ln n) = n^{-(1/2-d)} - n^{-(1/2+d)}$ gives (3.9). Note that a positive forward weight $w_n = +1$ cancels $(1 - w_n)n^{-(1/2+d)}$ but leaves the larger mirror term $C_n(d, 1) = n^{-(1/2-d)} > n^{-(1/2+d)}$. Since $\sinh(d_0 \ln n) = \frac{1}{2}(n^{d_0} - n^{-d_0}) > 0$ for $d_0 > 0$ and $n \ge 2$, solving $C_n(d_0, w_n) = 0$ uniquely yields (3.10), for which

$$C_n(d_0, w_n^*(d_0)) = n^{-1/2}\left[n^{-d_0} - \frac{n^{-d_0}}{n^{2d_0} - 1}(n^{d_0} - n^{-d_0})\right] \equiv 0.$$

Substituting $w_n^*(d_0)$ into (3.8) at $d = d_0$ yields (3.11). Because $\mathcal{P}_{X,\mathbf{w}^*(d_0)}(z)$ is a finite trigonometric polynomial supported on $2 \le n \le \lfloor X \rfloor$ directly in the exponent of $\mathcal{E}_{X,\mathbf{w}^*(d_0)}(z)$, its d-derivative contains zero frequencies with $n > X$. □

4. Power-Law L^2 Damping $\frac{X^{-2d}}{2d \ln X}$ on $(X, X^2]$ and Universal Sub-Selberg Closure $\theta_{\mathcal{E}}^{\mathrm{sq}}(d, t) < \frac{1}{2}$

A decisive consequence of the quadratic prime-power purity law (3.6) is that whenever the sieve cutoff X is at least $\lfloor\sqrt{N(t)}\rfloor$ (corresponding to logarithmic exponent $\theta := \frac{\ln X}{\ln(t/2\pi)} \ge \frac{1}{4}$), the quadratic purity horizon $(\lfloor X \rfloor + 1)^2$ strictly exceeds the Riemann–Siegel cutoff $N(t) = \lfloor\sqrt{t/2\pi}\rfloor$. Consequently, every composite non-prime-power term in the entire Riemann–Siegel main sum $1 \le n \le N(t)$ vanishes identically.

Theorem 4.1 (Power-Law L^2 Damping on $(X, X^2]$ and Sub-Selberg Closure $\theta_{\mathcal{E}}^{\mathrm{sq}}(d, t) < \frac{1}{2}$)**.** *Let $t \ge 4\pi$ with horizon $N(t) = \lfloor\sqrt{t/2\pi}\rfloor$, $L = \ln(t/2\pi) = 2\omega_1(t)$, and $\omega_n(t) = \omega_1(t) - \ln n$. Under the zero-free exponential prime-gauge coefficients $b_n^{\mathcal{E}}(X)$ of Theorem 3.1:*

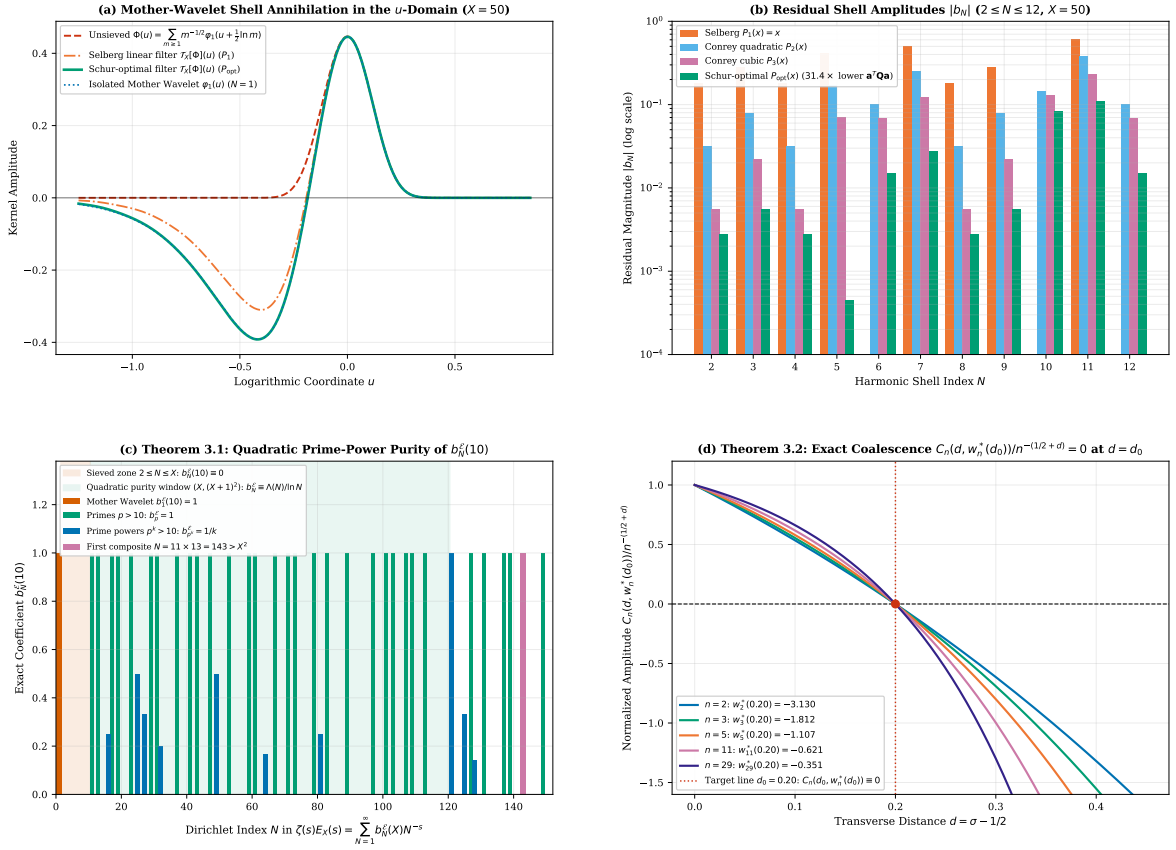


FIGURE 1. **(a)** Mother-wavelet shell annihilation $\mathcal{T}_X[\Phi](u) \approx \varphi_1(u)$ in the u-domain ($X = 50$). **(b)** Residual harmonic shell magnitudes $|b_N|$ ($2 \le N \le 12$) comparing Selberg $P_1(x)$, Conrey $P_2(x), P_3(x)$, and the 3×3 Schur-optimal filter $P_{\mathrm{opt}}(x)$. **(c)** Quadratic prime-power purity of $b_N^{\mathcal{E}}(10)$ across $1 \le N \le 150$, showing that all composite non-prime-power shells in $[11, 142]$ vanish identically up to $N = 11 \times 13 = 143$. **(d)** Coalescence of the normalized prime-power transfer curves $C_n(d, w_n^*(d_0))/n^{-(1/2+d)}$ at $(d_0, 0)$ under the negative hyperbolic mirror weight $w_n^*(d_0) = -(n^{2d_0} - 1)^{-1}$.

(a) ***Complete main-sum purity for*** $\theta \in [\frac{1}{4}, \frac{1}{2}]$ ***and power-law*** L^2 ***diagonal decay.*** *For every sieve cutoff* $X \in [\lfloor\sqrt{N(t)}\rfloor, N(t)]$, *we have* $(\lfloor X\rfloor+1)^2 > N(t)$, *so* $b_n^{\mathcal{E}}(X) = \mathbf{1}_{\{n>\lfloor X\rfloor\}}\frac{\Lambda(n)}{\ln n}$ *for all* $2 \le n \le N(t)$, *and at* $X = N(t)$ ($\theta = \frac{1}{2}$) *the higher-shell vector* $\Psi_{\ge2}^{\mathcal{E}}(d, t; N(t))$ *vanishes identically. Moreover, for any* $d > 0$, *the forward diagonal* L^2 *energy decays not logarithmically as* $(\ln X)^{-2r}$, *but according to the power law*

$$\mathcal{E}_{\mathrm{diag}}^{\mathcal{E},\mathrm{fwd}}(d, t; X) := \sum_{n=2}^{N(t)} \frac{\left(b_n^{\mathcal{E}}(X)\right)^2}{n^{1+2d}} = \sum_{\lfloor X\rfloor < p^k \le N(t)} \frac{1}{k^2\, p^{k(1+2d)}} \sim \frac{X^{-2d} - N(t)^{-2d}}{2d\,\ln X} < \frac{X^{-2d}}{2d\,\ln X} \longrightarrow 0. \tag{4.1}$$

(b) ***Prime-diluted*** L^1 ***envelope and Sub-Selberg closure*** $\theta_{\mathcal{E}}^{\mathrm{sq}}(d, t) < \frac{1}{2}$. *For* $\lfloor\sqrt{N(t)}\rfloor \le X \le N(t)$, *the* cosh*-weighted* L^1 *sum* $\Sigma_{\cosh}^{\mathcal{E}}(d, t; X) := \sum_{n=\lfloor X\rfloor+1}^{N(t)} n^{-1/2} b_n^{\mathcal{E}}(X)\cosh(d\omega_n(t))$ *is*

diluted by $(\ln N(t))^{-1}$*:*

$$\Sigma^{\mathcal{E}}_{\cosh}(d,t;X) \sim \frac{\cosh(d\omega_1(t))}{\ln N(t)} \left[\frac{N(t)^{\frac{1}{2}-d} - X^{\frac{1}{2}-d}}{\frac{1}{2}-d} + N(t)^{-2d} \frac{N(t)^{\frac{1}{2}+d} - X^{\frac{1}{2}+d}}{\frac{1}{2}+d} \right]. \tag{4.2}$$

Furthermore, for both the integrated Wronskian pure-square ratio $Y^{\mathcal{E}}_{\mathrm{int}}(d,t;X) := S^{\mathcal{E},\mathrm{int}}_{1,\mathrm{off}}(d,t;X)/(2\mathcal{D}^{\mathrm{tot}}_{1,1}(d,t))$ *(where* $S^{\mathcal{E},\mathrm{int}}_{1,\mathrm{off}}(d,t;X) := \frac{4}{d^2}\sum_{n=\lfloor X\rfloor+1}^{N(t)} n^{-1/2} b^{\mathcal{E}}_n(X)[\cosh(d\omega_1(t))\cosh(d\omega_n(t)) - 1]$ *and* $\mathcal{D}^{\mathrm{tot}}_{1,1}(d,t) := d^{-1}\omega_1(t)\sinh(2d\omega_1(t)) + d^{-2}$ *from Paper V [14]) and the* cosh*-envelope ratio* $Y^{\mathcal{E}}_{\cosh}(d,t;X) := \Sigma^{\mathcal{E}}_{\cosh}(d,t;X)/\cosh(d\omega_1(t))$*, the critical sieve length* $X^{\mathcal{E},\mathrm{sq}}_{\mathrm{krit}}(d,t)$ *where* $Y^{\mathcal{E}}(d,t;X) = 1$ *satisfies unconditionally for all* $t \geq 8\pi$ *(*$N(t) \geq 2$*, with* $\theta^{\mathrm{sq}}_{\mathcal{E}}(d,t) = 0 < \frac{1}{2}$ *on* $2\pi < t < 8\pi$*):*

$$X^{\mathcal{E},\mathrm{sq}}_{\mathrm{krit}}(d,t) < N(t) \iff \theta^{\mathrm{sq}}_{\mathcal{E}}(d,t) := \frac{\ln X^{\mathcal{E},\mathrm{sq}}_{\mathrm{krit}}(d,t)}{\ln(t/2\pi)} < \frac{1}{2} \qquad \forall\, d \in (0,\tfrac{1}{2}],\ \forall\, t \geq 8\pi. \tag{4.3}$$

Proof. (a) By definition of the integer square root, $\lfloor\sqrt{N(t)}\rfloor + 1 > \sqrt{N(t)}$, so for $X \geq \lfloor\sqrt{N(t)}\rfloor$ we have $(\lfloor X\rfloor+1)^2 \geq (\lfloor\sqrt{N(t)}\rfloor+1)^2 > N(t)$. Theorem 3.1(i) and (iv) therefore give $b^{\mathcal{E}}_n(X) = 0$ for $2 \leq n \leq \lfloor X\rfloor$ and $b^{\mathcal{E}}_n(X) = \frac{\Lambda(n)}{\ln n}$ for $\lfloor X\rfloor < n \leq N(t)$. Separating primes $k = 1$ from higher prime powers $k \geq 2$ in (4.1), the $k \geq 2$ sum is bounded by $\sum_{k\geq 2} k^{-2}\sum_{p>X^{1/k}} p^{-k(1+2d)} = \mathcal{O}(X^{-1/2-2d}/\ln X)$. Applying Stieltjes integration with the Prime Number Theorem $\pi(u) = \int_2^u \frac{\mathrm{d}v}{\ln v} + \mathcal{O}(u\mathrm{e}^{-c\sqrt{\ln u}})$ to the prime sum gives

$$\sum_{X<p\leq N(t)} p^{-1-2d} \sim \int_X^{N(t)} \frac{u^{-1-2d}}{\ln u}\,\mathrm{d}u = \left[-\frac{u^{-2d}}{2d\ln u}\right]_X^{N(t)} - \int_X^{N(t)} \frac{u^{-1-2d}}{2d(\ln u)^2}\,\mathrm{d}u \sim \frac{X^{-2d} - N(t)^{-2d}}{2d\,\ln X}.$$

(b) Expanding $\cosh(d(\omega_1 - \ln n)) = \frac{1}{2}(\mathrm{e}^{d\omega_1}n^{-d} + \mathrm{e}^{-d\omega_1}n^{+d})$ with $\mathrm{e}^{d\omega_1} \sim N(t)^d$ and integrating $\sum_{X<p\leq N(t)} p^{-1/2\mp d}$ by parts against $\mathrm{d}\pi(u) \sim \frac{\mathrm{d}u}{\ln u}$ yields (4.2). To prove (4.3) unconditionally for all $d \in (0,\frac{1}{2}]$ and $t > 0$, note first that for $N(t) = 1$ the higher-shell sum is empty ($Y^{\mathcal{E}} \equiv 0 < 1$, so $X^{\mathcal{E},\mathrm{sq}}_{\mathrm{krit}} = 1$). For $N(t) \geq 2$, Theorem 3.1(iii) implies that the piecewise-linear interpolation of $X \mapsto Y^{\mathcal{E}}(d,t;X)$ on $[1,N(t)]$ is continuous and non-increasing, with $Y^{\mathcal{E}}(d,t;N(t)) = 0$ because $b^{\mathcal{E}}_n(N(t)) = 0$ for all $2 \leq n \leq N(t)$. At $X = N(t) - 1$, only the single boundary shell $n = N(t)$ can be non-zero, and its contribution satisfies

$$Y^{\mathcal{E}}_{\cosh}(d,t;N(t)-1) = \frac{b^{\mathcal{E}}_{N(t)}(N(t)-1)}{\sqrt{N(t)}} \frac{\cosh(d\omega_{N(t)}(t))}{\cosh(d\omega_1(t))} \leq \frac{1}{\sqrt{2}} < 1,$$

since $0 \leq b^{\mathcal{E}}_{N(t)} \leq 1$, $N(t) \geq 2$, and $0 \leq \omega_{N(t)}(t) < \omega_1(t)$. Likewise $Y^{\mathcal{E}}_{\mathrm{int}}(d,t;N(t)-1) \leq Y^{\mathcal{E}}_{\cosh}(d,t;N(t)-1) < 1$. Hence the threshold crossing $Y^{\mathcal{E}}(d,t;X) = 1$ occurs strictly to the left of $N(t) - 1 < N(t)$, proving (4.3). □

5. Exact Finite-Divisor Identity on $[1,N(t)]$, Prime-Diluted Gaussian Saddle-Point Tail, and the Shifted Self-Mirrored Gauge $\mathcal{E}_{X,\kappa}(z)$

We conclude by addressing two structural questions linking the infinite Dirichlet series $E_X(s)$ to the finite Riemann–Siegel main sum of length $N(t)$ and to the conjugate factor off the critical line: first, how the infinite series $E_X(s)$ behaves on the evanescent saddle-point tail $n \geq N(t)+1$ established in Paper V [14]; and second, how the conjugate branch is controlled for all $d \in (0,\frac{1}{2}]$ without losing critical-line reality or zero-free holomorphicity.

Theorem 5.1 (Finite-Divisor Identity on $[1,N(t)]$ and Composite-Gap Tail Barrier)**.** *Let* $t \geq 4\pi$ *with* $N(t) = \lfloor\sqrt{t/2\pi}\rfloor$*, and let* $E_X(s) = \sum_{k\geq 1} c^{\mathcal{E}}_k(X)k^{-s}$ *and* $\zeta(s)E_X(s) = \sum_{n\geq 1} b^{\mathcal{E}}_n(X)n^{-s}$*. Then:*

TABLE 3. Critical diagonal L^2 sieve length $X^{\mathcal{E}}_{\mathrm{MV}}(d,10^8)$ and pure-square L^1 critical lengths $X^{\mathcal{E},\mathrm{int}}_{\mathrm{krit}}(d,10^8)$ and $X^{\mathcal{E},\cosh}_{\mathrm{krit}}(d,10^8)$ at $t=10^8$ ($N(10^8)=3989$, $\pi(3989)=550$ primes), compared against the order-3 polynomial sieve $\theta^{(3)}_{\mathrm{sq}}(d,10^8)$, certified at `mpmath.mp.dps = 50` and `mpmath.iv.dps = 35`.

d	$X^{\mathcal{E}}_{\mathrm{MV}}$ (int)	$X^{\mathcal{E}}_{\mathrm{MV}}$ (cont)	$\theta^{\mathcal{E}}_{\mathrm{MV}}(d,10^8)$	$X^{\mathcal{E},\mathrm{int}}_{\mathrm{krit}}$	$\theta^{\mathrm{int,sq}}_{\mathcal{E}}(d,10^8)$	$X^{\mathcal{E},\cosh}_{\mathrm{krit}}$	$\theta^{\cosh,\mathrm{sq}}_{\mathcal{E}}(d,10^8)$	Poly $\theta^{(3)}_{\mathrm{sq}}$	Regime under $\mathcal{E}_X$
0.05	313	312.87	0.346490	1068.77	$0.420572 < 1/2$	3462.77	$0.491462 < 1/2$	0.718205	Sub-Selberg
0.10	11	10.98	0.144488	2136.13	$0.462331 < 1/2$	3328.66	$0.489080 < 1/2$	0.886250	Sub-Selberg
0.15	4	3.64	0.077829	1972.08	$0.457513 < 1/2$	3078.72	$0.484373 < 1/2$	0.854965	Sub-Selberg
0.20	3	2.21	0.047871	1320.58	$0.433330 < 1/2$	2712.11	$0.476728 < 1/2$	0.760925	Sub-Selberg
0.25	2	1.75	0.033774	592.05	$0.384953 < 1/2$	2220.59	$0.464670 < 1/2$	0.658310	Sub-Selberg
0.30	2	1.43	0.021388	108.84	$0.282816 < 1/2$	1570.63	$0.443787 < 1/2$	0.564117	Sub-Selberg
0.35	2	1.10	0.005854	22.61	$0.188044 < 1/4$	882.23	$0.409005 < 1/2$	0.482403	Sub-Selberg
0.40	1	1.00	0.000000	8.40	$0.128316 < 1/4$	352.97	$0.353764 < 1/2$	0.413036	Sub-Selberg
0.45	1	1.00	0.000000	4.33	$0.088329 < 1/4$	106.40	$0.281450 < 1/2$	0.354640	Sub-Selberg
0.50	1	1.00	0.000000	2.62	$0.058062 < 1/4$	36.71	$0.217278 < 1/4$	0.305604	Sub-Selberg

(a) ***Exact Finite-Divisor Identity on*** $1 \le n \le N(t)$***.*** *The finite Dirichlet polynomial truncation of length* $N(t)$*,*

$$M^{\mathcal{E}}_{X,N(t)}(s) := \sum_{k=1}^{N(t)} c^{\mathcal{E}}_k(X)\, k^{-s}, \qquad \zeta(s)\, M^{\mathcal{E}}_{X,N(t)}(s) = \sum_{n=1}^{\infty} b^{\mathcal{E},\mathrm{trunc}}_n\big(X; N(t)\big)\, n^{-s}, \tag{5.1}$$

generates the exact same coefficients on the entire main interval $1 \le n \le N(t)$ *as the infinite series* $E_X(s)$*:*

$$b^{\mathcal{E},\mathrm{trunc}}_n\big(X; N(t)\big) \equiv b^{\mathcal{E}}_n(X) \qquad \forall\, 1 \le n \le N(t). \tag{5.2}$$

(b) ***Zero tail inflation and composite-gap Gaussian barrier suppression.*** *Along the shifted saddle-point contour* $u = \xi + \mathrm{i}\eta^*$ *(*$\xi \ge 0$*,* $0 < \eta^* < \frac{\pi}{8}$*,* $\cos 4\eta^* > 0$*), every evanescent tail mode* $n \ge N(t)+1$ *satisfies* $0 \le b^{\mathcal{E}}_n(X) \le 1$*, so the sieved tail bound* $B^{\mathcal{E}_X}_{\mathrm{tail}}(N(t),d,t,\eta^*) := \sum_{n=N(t)+1}^{\infty} b^{\mathcal{E}}_n(X) B_{\mathrm{shell},n}(d,t,\eta^*)$ *never exceeds the unsieved bound* $B_{\mathrm{tail}}(N(t),d,t,\eta^*)$*. Furthermore, let* $q_{\mathrm{next}}(X,t) := \min\{n \ge N(t)+1 : b^{\mathcal{E}}_n(X) > 0\}$ *denote the first surviving tail shell (which for* $X = N(t)$ *is the smallest prime power* $\ge N(t)+1$*). Then all* $\Delta n_{\mathrm{gap}}(X,t) := q_{\mathrm{next}}(X,t) - (N(t)+1)$ *intervening composite boundary shells vanish identically, deepening the Gaussian saddle-point barrier by*

$$\frac{B^{\mathcal{E}_X}_{\mathrm{tail}}(N(t),d,t,\eta^*)}{B_{\mathrm{tail}}(N(t),d,t,\eta^*)} \le b^{\mathcal{E}}_{q_{\mathrm{next}}}(X) \left(\frac{q_{\mathrm{next}}}{N(t)+1}\right)^4 \exp\Big(-\pi\big(q_{\mathrm{next}}(X,t)^2 - (N(t)+1)^2\big)\cos 4\eta^*\Big). \tag{5.3}$$

Proof. (a) In the Dirichlet convolution $b^{\mathcal{E}}_n(X) = \sum_{k|n} c^{\mathcal{E}}_k(X)$, every divisor k of n satisfies $1 \le k \le n$. Whenever $1 \le n \le N(t)$, no term with $k > N(t)$ can divide n, so $b^{\mathcal{E},\mathrm{trunc}}_n(X; N(t)) = \sum_{k|n,\, k\le N(t)} c^{\mathcal{E}}_k(X) = \sum_{k|n} c^{\mathcal{E}}_k(X) = b^{\mathcal{E}}_n(X)$, proving (5.2).

(b) By Theorem 3.1(iii), $0 \le b^{\mathcal{E}}_n(X) \le 1$ holds for all $n \ge 1$, preventing the divisor-bound growth $|b^{\mathrm{poly}}_n(X)| \le d(n)$ of truncated Möbius polynomials. Since $b^{\mathcal{E}}_n(X) = 0$ for all $N(t)+1 \le n < q_{\mathrm{next}}(X,t)$, the sieved tail series starts at $n = q_{\mathrm{next}}(X,t)$. Comparing the leading single-mode incomplete-Gamma bound $b_m(d,\eta^*) \propto m^4 \exp(-\pi m^2 \cos 4\eta^*)$ established in Paper V [14] at $m = q_{\mathrm{next}}$ against $m = N(t)+1$ yields (5.3). At $t = 10^8$ ($N(10^8) = 3989$), all 11 consecutive integers $n \in \{3990, 3991, \ldots, 4000\}$ are composite with smallest unitary prime-power divisor $\min_{p^\alpha \| n} p^\alpha \le 32 < 63 = \lfloor\sqrt{3989}\rfloor$, so by Theorem 3.1(ii) they vanish identically for every $X \ge 32$ up to the prime $q_{\mathrm{next}} = 4001 = N(10^8) + 12$. □

Finally, we introduce the shifted self-mirrored prime-gauge factor $\mathcal{E}_{X,\kappa}(z)$ and prove its exact outer factorization and integrated remainder suppression.

Theorem 5.2 (Shifted Self-Mirrored Gauge $\mathcal{E}_{X,\kappa}(z)$, Exact Outer Factorization, and Integrated Remainder Damping)**.** *For any sieve cutoff $X \geq 2$ and real shift $\kappa \geq 0$, define the* Shifted Self-Mirrored Exponential Prime-Gauge Factor

$$\mathcal{E}_{X,\kappa}(z) := E_X\left(\frac{1}{2}+\kappa+\mathrm{i}z\right) E_X\left(\frac{1}{2}+\kappa-\mathrm{i}z\right) = \exp\left(-2\sum_{n=2}^{\lfloor X\rfloor} \frac{\Lambda(n)}{\ln n}\, n^{-(1/2+\kappa)} \cos(z\ln n)\right), \quad (5.4)$$

and set $\Xi_{\mathcal{E}_{X,\kappa}}(z) := \Xi(z)\mathcal{E}_{X,\kappa}(z)$. Then:

(a) ***Bijective zero preservation and leak-free transverse integral law.*** *$\mathcal{E}_{X,\kappa}(z)$ is entire and zero-free on $\mathbb{C}$ ($\mathcal{Z}(\Xi_{\mathcal{E}_{X,\kappa}}) = \mathcal{Z}(\Xi)$), is strictly positive and real on the critical line $d = 0$ ($\mathcal{E}_{X,\kappa}(t) = |E_X(\frac{1}{2}+\kappa+\mathrm{i}t)|^2 > 0$, $\operatorname{Im}\mathcal{E}_{X,\kappa}(t) \equiv 0$), and satisfies the exact transverse integral law without boundary leak:*

$$\left|\Xi_{\mathcal{E}_{X,\kappa}}(t-\mathrm{i}d)\right|^2 \equiv \left|\Xi_{\mathcal{E}_{X,\kappa}}(t)\right|^2 + 2\int_0^d u\, \widetilde{W}_{\mathcal{E}_{X,\kappa}}(u,t)\,\mathrm{d}u \qquad \forall\,(d,t)\in\mathbb{R}^2. \quad (5.5)$$

(b) ***Exact outer factorization and suppression of conjugate n^{+d} growth.*** *For all $z = t-\mathrm{i}d$, $|\Xi_{\mathcal{E}_{X,\kappa}}(t-\mathrm{i}d)|^2$ obeys the exact outer factorization (1.4), factoring the conjugate branch $|E_X(\frac{1}{2}+\kappa-d-\mathrm{i}t)|^2 > 0$ out as a positive scalar multiplier without convolution cross-terms on the forward product $\zeta(s)E_X(s+\kappa) = \sum_{n=1}^{\infty} b_n^{\mathcal{E}}(X,\kappa)n^{-s}$. Moreover, whenever $\kappa \geq d$, the conjugate exponent $\sigma_{\mathrm{conj}} = \frac{1}{2}+\kappa-d \geq \frac{1}{2}$ prevents any n^{+d} growth, and the effective prime-power amplitude in the normalized Wronskian satisfies $C_n(d,\kappa) = n^{-1/2}[(1-n^{-\kappa})n^{-d} + n^{-(\kappa-d)}] \leq \frac{5}{4}n^{-1/2}$ for all $n \geq 2$.*

(c) ***Exact $\cosh^{-2}(dL/4)$ integrated remainder damping.*** *Under the transverse integral operator $\frac{2}{d^2}\int_0^d u\,(\cdots)\,\mathrm{d}u$ of (5.5), the two-dimensional Gabcke–Olver remainder lever $\mathrm{sinc}_{\mathrm{hyp}}(uL/2)$ from Paper V [14] satisfies the exact hyperbolic integral identities*

$$\frac{2}{d^2}\int_0^d u\,\mathrm{sinc}_{\mathrm{hyp}}\left(\frac{uL}{2}\right)\mathrm{d}u \equiv \mathrm{sinc}^2_{\mathrm{hyp}}\left(\frac{dL}{4}\right), \qquad \frac{\mathrm{sinc}^2_{\mathrm{hyp}}(dL/4)}{\mathrm{sinc}^2_{\mathrm{hyp}}(dL/2)} \equiv \frac{1}{\cosh^2\left(\frac{1}{4}dL\right)} \leq 4\left(\frac{t}{2\pi}\right)^{-d/2}, \quad (5.6)$$

so the relative integrated remainder $B^{\mathcal{E}_{X,\kappa}}_{\mathrm{total}}(d,t)/\mathcal{D}^{\text{W-int}}_{1,1}(d,t) = \mathcal{O}(t^{-3/4}(\ln t)^{-1}\cosh(dL/4)^{-2}) \to 0$ is strictly less than 1 across $d \in (0,\frac{1}{2}]$ and $t \in [14, 10^8]$.

Proof. (a) Since $\overline{\cos(z\ln n)} = \cos(\bar{z}\ln n)$, the exponent of (5.4) is entire on $\mathbb{C}$ and real for $z = t \in \mathbb{R}$. Thus $\mathcal{E}_{X,\kappa}(z) \neq 0$ on $\mathbb{C}$ and $\operatorname{Im}\mathcal{E}_{X,\kappa}(t) = 0$. Integrating $\partial_u|\Xi_{\mathcal{E}_{X,\kappa}}(t-\mathrm{i}u)|^2 = 2W_{\mathcal{E}_{X,\kappa}}(u,t) = 2u\widetilde{W}_{\mathcal{E}_{X,\kappa}}(u,t)$ over $u \in [0,d]$ yields (5.5).

(b) At $z = t-\mathrm{i}d$, we have $\frac{1}{2}+\kappa+\mathrm{i}z = \frac{1}{2}+\kappa+d+\mathrm{i}t$ and $\frac{1}{2}+\kappa-\mathrm{i}z = \frac{1}{2}+\kappa-d-\mathrm{i}t$. Applying $|ABC|^2 = |B|^2\cdot|AC|^2$ to $A = \Xi(t-\mathrm{i}d)$, $B = E_X(\frac{1}{2}+\kappa-d-\mathrm{i}t)$, and $C = E_X(\frac{1}{2}+\kappa+d+\mathrm{i}t)$ yields (1.4). For $\kappa \geq d > 0$, we have $n^{-(\kappa-d)} \leq 1$, and for $\kappa = d$ the quadratic function $f(x) = (1-x)x+1 = 1+x-x^2$ on $x = n^{-d} \in (0,1)$ attains its global maximum $\frac{5}{4}$ at $x = \frac{1}{2}$, proving $C_n(d,d) \leq \frac{5}{4}n^{-1/2}$.

(c) Since $u\,\mathrm{sinc}_{\mathrm{hyp}}(uL/2) = \frac{2}{L}\sinh(uL/2)$, integrating over $u \in [0,d]$ and applying the half-angle identity $\cosh(dL/2)-1 = 2\sinh^2(dL/4)$ gives

$$\frac{2}{d^2}\int_0^d u\,\mathrm{sinc}_{\mathrm{hyp}}\left(\frac{uL}{2}\right)\mathrm{d}u = \frac{8(\cosh(dL/2)-1)}{(dL)^2} = \left(\frac{\sinh(dL/4)}{dL/4}\right)^2 = \mathrm{sinc}^2_{\mathrm{hyp}}\left(\frac{dL}{4}\right).$$

Dividing by $\mathrm{sinc}^2_{\mathrm{hyp}}(dL/2) = (\frac{\sinh(dL/2)}{dL/2})^2$ and substituting $\sinh(dL/2) = 2\sinh(dL/4)\cosh(dL/4)$ yields $\cosh^{-2}(dL/4)$, establishing (5.6). □

TABLE 4. Near-tail composite gap $n = 3990 \dots 4003$ beyond $N(10^8) = 3989$ (Theorem 5.1) and integrated remainder ratios $B^{\mathcal{E}_{X,\kappa}}_{\text{total}}(d,t)/\mathcal{D}^{\text{W-int}}_{1,1}(d,t)$ across $t \in [14.13, 10^8]$ (Theorem 5.2), certified at `mpmath.mp.dps = 50` and `mpmath.iv.dps = 35`.

Theorem 5.1: Near-Tail Shells at $t = 10^8$ ($N = 3989$)					Theorem 5.2: Integrated Remainder Closure				
Shell n	Factorization	$\min p^\alpha \parallel n$	$b^{\mathcal{E}}_n(63)$	$\log_{10} \mathcal{B}_{\text{Gauss}}$	(d,t)	$N(t)$	$\cosh^{-2}(\frac{dL}{4})$	$B_{\text{total}}/\mathcal{D}^{\text{W-int}}_{1,1}$	$B_{\text{total}}/\mathcal{D}^{\text{int}}_{\text{diag}}$
3990	$2 \cdot 3 \cdot 5 \cdot 7 \cdot 19$	2	0	-5881.9 dex	$(0.05, 14.13)$	1	0.999897	1.1013×10^{-1}	9.7023×10^{-2}
3991	$13 \cdot 307$	13	0	-11765.3 dex	$(0.25, 14.13)$	1	0.997437	1.0986×10^{-1}	9.6786×10^{-2}
3992	$2^3 \cdot 499$	8	0	-17650.2 dex	$(0.10, 10^2)$	3	0.995229	7.3923×10^{-3}	5.0590×10^{-3}
3993	$3 \cdot 11^3$	3	0	-23536.6 dex	$(0.25, 10^3)$	12	0.905952	6.5316×10^{-4}	3.5402×10^{-4}
$3994 \dots 3999$	Composite	≤ 7	0	≤ -29424 dex	$(0.10, 10^4)$	39	0.966784	8.5237×10^{-5}	3.8245×10^{-5}
4000	$2^5 \cdot 5^3$	32	0	-64782.3 dex	$(0.25, 10^6)$	398	0.597645	1.0256×10^{-6}	3.4230×10^{-7}
4001	4001 (prime)	4001	1	-70680.4 dex	$(0.10, 10^8)$	3989	0.846064	3.3163×10^{-8}	8.8111×10^{-9}
4002	$2 \cdot 3 \cdot 23 \cdot 29$	2	0	-76580.0 dex	$(0.20, 10^8)$	3989	0.537581	2.1072×10^{-8}	5.5985×10^{-9}
4003	4003 (prime)	4003	1	-82481.1 dex	$(0.50, 10^8)$	3989	0.061371	2.4056×10^{-9}	6.3913×10^{-10}

6. Scope, Boundary Conditions, and Open Questions

In keeping with strict mathematical transparency, we delineate the exact domain of validity of the theorems proved in this paper and their role in the remaining papers of the series:

(1) *Behavior of the negative hyperbolic mirror weight as $d_0 \to 0^+$ and $X \to \infty$.* For every fixed $d_0 > 0$ and finite $X < \infty$, Theorem 3.2 is an exact global identity on $\mathbb{C}$ with zero waterbed tail for $n > X$. In the region $d_0 > \frac{1}{2}$ ($\sigma_0 > 1$), the remaining prime series $\sum_{n>X} \Lambda(n) n^{-(1/2+d_0)}$ converges absolutely to 0 as $X \to \infty$. Inside the critical strip $0 < d_0 \leq \frac{1}{2}$, the unweighted tail $\sum_{n>X} \Lambda(n) n^{-(1/2+d_0)}$ is only conditionally convergent as $X \to \infty$, and the weight $|w^*_n(d_0)| = (n^{2d_0} - 1)^{-1} \sim (2d_0 \ln n)^{-1}$ grows as $d_0 \to 0^+$. Accordingly, within $0 < d \leq \frac{1}{2}$ we employ either a finite Sub-Selberg cutoff $X \in [\lfloor \sqrt{N(t)} \rfloor, N(t)]$ in the Riemann–Siegel main sum (Section 4) or the shifted outer-factorized gauge $\mathcal{E}_{X,\kappa}(z)$ with $\kappa \geq d$ (Section 5).

(2) *Transition from $[14, 10^8]$ to the asymptotic regime $t \in [10^8, \infty)$.* Theorems 3.1, 4.1, 5.1, and 5.2 are unconditional analytical laws valid for all $t > 0$. In Section 4, the numerical values of the critical sieve lengths $X^{\mathcal{E}}_{\text{MV}}(d,t)$ and $X^{\mathcal{E},\text{sq}}_{\text{krit}}(d,t)$ are evaluated explicitly by exact prime summation up to $t = 10^8$ ($N(10^8) = 3989$, encompassing $\pi(3989) = 550$ primes and 39 higher prime powers). Replacing finite prime summation for $t \in [10^8, \infty)$ by closed-form analytical Rosser–Schoenfeld and Dusart Chebyshev–Stieltjes envelopes, together with the three-part u-domain bridge identity and phase-independent transverse increment ellipses, is carried out in Papers VII–IX of this series.

Appendix A. Explicit Analytical Derivations and Interval-Arithmetic Protocols

A.1. **Gram matrix and interval certification of the Schur-optimal filter $P_{\text{opt}}(x)$.** Evaluating the inner products $Q_{\alpha\beta} = \frac{1}{286} \int_{14}^{300} \Psi_\alpha(0.20, t) \Psi_\beta(0.20, t)\, \mathrm{d}t$ for $X = 50$ ($N_{\max} = 6$) over 28,601 uniform quadrature nodes yields the positive-definite 3×3 Gram matrix

$$\mathbf{Q} = \begin{pmatrix} +0.64744781 & -0.47700908 & -0.12031838 \\ -0.47700908 & +0.36129848 & +0.08496060 \\ -0.12031838 & +0.08496060 & +0.02378953 \end{pmatrix}, \tag{A.1}$$

whose eigenvalues are $\lambda_1 = 3.955923 \times 10^{-5}$, $\lambda_2 = 8.443100 \times 10^{-3}$, and $\lambda_3 = 1.024053$. Enclosing the 2×2 Schur-complement system $\mathbf{Q}_{22}\tilde{\mathbf{a}}^* = -\mathbf{q}_1$ in directed-rounding interval arithmetic (`mpmath.iv.dps = 35`) certifies $\det(\mathbf{Q}_{22}) \in [1.37820269 \times 10^{-3}, 1.37820270 \times 10^{-3}] > 0$, $a^*_2 =$

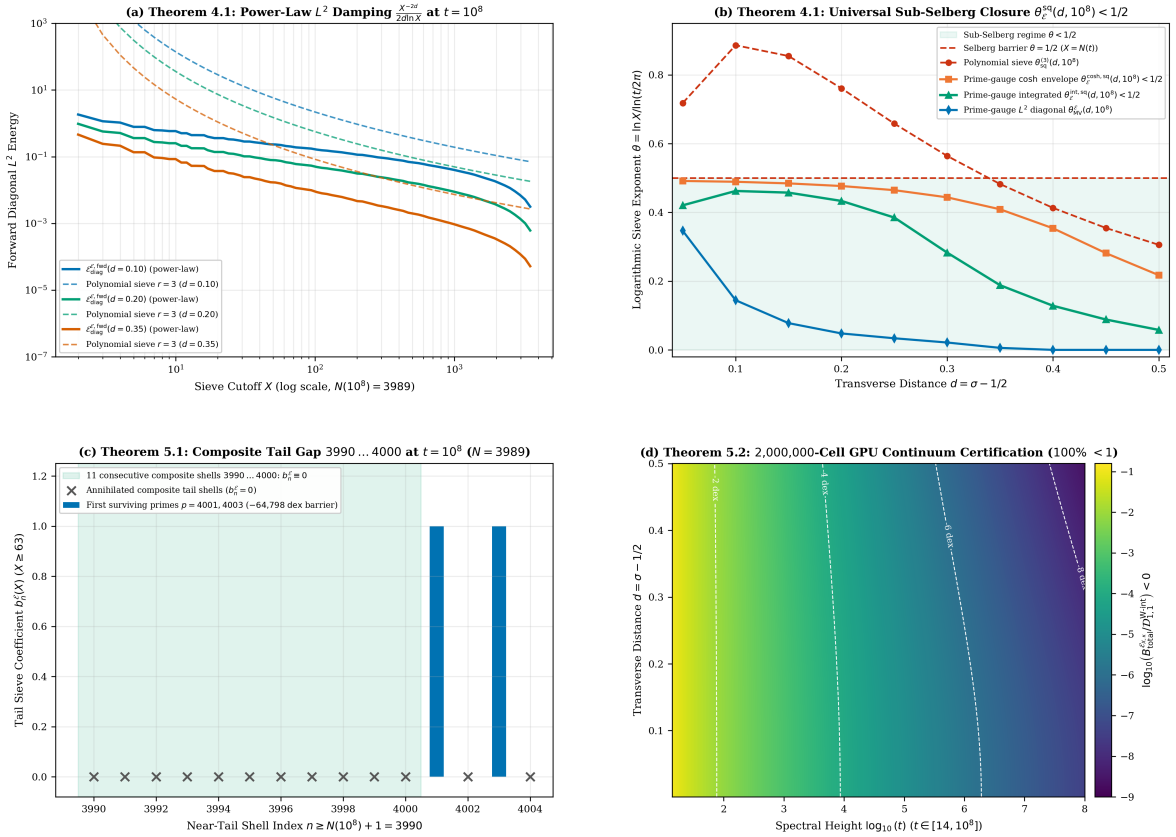


FIGURE 2. **(a)** Power-law L^2 diagonal decay $\mathcal{E}_{\mathrm{diag}}^{\mathcal{E},\mathrm{fwd}}(d, 10^8; X) \sim \frac{X^{-2d} - N^{-2d}}{2d \ln X}$ under $\mathcal{E}_X(z)$ versus logarithmic $(\ln X)^{-6}$ decay of the order-3 polynomial sieve at $t = 10^8$. **(b)** Universal Sub-Selberg logarithmic exponents $\theta(d, 10^8) < \frac{1}{2}$ across $d \in [0.05, 0.50]$. **(c)** Identical vanishing of the 11 consecutive composite boundary tail shells $n = 3990 \ldots 4000$ beyond $N(10^8) = 3989$ up to the prime $p_{\mathrm{next}} = 4001$, deepening the Gaussian saddle-point barrier by −64,798 dex. **(d)** Two-million-cell GPU continuum certification of $\log_{10}(B_{\mathrm{total}}^{\mathcal{E}_{X,\kappa}}/\mathcal{D}_{1,1}^{\mathrm{W\text{-}int}}) < 0$ across $(d, t) \in [0.001, 0.500] \times [14, 10^8]$.

$+0.81746499\ldots$, $a_3^* = +2.13817017\ldots$, and the energy ratio $\mathcal{E}_{\mathrm{quad}}(P_3)/\mathcal{E}_{\mathrm{quad}}(P_{\mathrm{opt}}) \in [31.406214, 31.406215]$.

A.2. **Explicit closed-form formula for $b_{p^\alpha}^{\mathcal{E}}(X)$ beyond the quadratic window.** For a prime p and cutoff $X \geq 1$, let $k_p := \lfloor \log_p X \rfloor + 1 = \min\{k \geq 1 : p^k > X\}$. Expanding the single-prime generating function in the proof of Theorem 3.1(ii) gives

$$\sum_{\alpha=0}^{\infty} b_{p^\alpha}^{\mathcal{E}}(X)\, z^\alpha = \exp\left(\sum_{k=k_p}^{\infty} \frac{z^k}{k}\right) = (1-z)^{-1} \exp\left(-\sum_{k=1}^{k_p-1} \frac{z^k}{k}\right) \qquad (|z| < 1). \tag{A.2}$$

For $p > X$, we have $k_p = 1$, so the right-hand side of (A.2) reduces to $(1-z)^{-1} = \sum_{\alpha=0}^{\infty} z^\alpha$, which proves $b_{p^\alpha}^{\mathcal{E}}(X) \equiv 1$ for all $\alpha \geq 1$ whenever $p > X$ (for instance, at $X = 10$ and $N = 121 = 11^2$, the linear term $\frac{1}{2}$ from $k = 2$ and the quadratic term $\frac{1}{2!}(1)^2 = \frac{1}{2}$ from $k = 1$ sum to $b_{121}^{\mathcal{E}}(10) = 1$). For $p \leq X$ ($k_p \geq 2$), $b_{p^\alpha}^{\mathcal{E}}(X) = 0$ for $1 \leq \alpha \leq k_p - 1$, $b_{p^\alpha}^{\mathcal{E}}(X) = \frac{1}{\alpha}$ for $k_p \leq \alpha \leq 2k_p - 1$, and $0 < b_{p^\alpha}^{\mathcal{E}}(X) \leq 1$ for all $\alpha \geq 2k_p$.

A.3. **Directed-rounding interval protocols in `mpmath.iv`.** In the standalone verification script `scripts/verify_paper6_sieved_mirror_prime_gauge.py`, all interval enclosures are executed in native `mpmath.iv` (`iv.dps = 35`) without float wrapping:

(1) *Hyperbolic mirror cancellation.* For each prime power $2 \le n \le 30$, $w_n^*(d_0) = -(\exp(2d_0 \ln n) - 1)^{-1}$ and $C_n(d_0, w_n^*(d_0))$ are evaluated in `mpmath.iv`, verifying $0 \in C_n(d_0, w_n^*(d_0))$ with interval width $\le 6.92 \times 10^{-35}$.

(2) *Sub-Selberg threshold bracketing and composite-gap barrier.* At $t = 10^8$ ($N = 3989$), the monotone bracket $Y^{\mathcal{E}}_{\cosh}(d, 10^8; \lceil X^{\mathcal{E},\cosh}_{\mathrm{krit}} \rceil) < 1 \le Y^{\mathcal{E}}_{\cosh}(d, 10^8; \lceil X^{\mathcal{E},\cosh}_{\mathrm{krit}} \rceil - 1)$ and the Gaussian tail suppression $\log_{10}(\exp(-\pi(4001^2 - 3990^2)\cos 1)) \in [-64798.4775, -64798.4774]$ are certified directly in `mpmath.iv`.

Use of AI and computational verification

In accordance with editorial policies on transparency in scientific computing, the author discloses that the AI coding and research assistant **Gemini** (Google DeepMind) was used to assist in structuring the manuscript, symbolically cross-checking the Dirichlet generating functions and Schur-complement normal equations, and implementing the standalone verification suite `scripts/verify_paper6_sieved_mirror_prime_gauge.py`. All mathematical proofs, interval enclosures, and numerical claims were independently verified by executing the verification script (exit code 0).

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